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Cyber Security

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QUANTIFYING COMMUNICATION DEGRADATION UNDER CYBERATTACKS IN SMARTGRIDSIM (SGSIM)

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Abstract

Traditional power grid systems are rapidly transitioning to modern smart grid power systems. Modern smart grids depend on real time communication networks for monitoring and control, which makes them vulnerable to cyberattacks that can degrade network performance and system reliability. This research paper studies how cyberattacks affect network performance in a smart grid by measuring latency, jitter, packet loss and inter-packet delay variation (IPDV). A simulated smart grid network was built using SmartGridSim and Mininet to model communication between smart grid devices. Network traffic was recorded during the baseline operation and during simulated attacks, including Denial of Service (DoS) and False Data Injection (FDI). Packet data was captured with Wireshark and later analyzed statistically. The results show the DoS scenario caused the clearest degradation in communication timing with greater variation in delay-related metrics than in the baseline condition, while the FDI scenario remained closer to normal operation in timing terms. Statistical summaries and visualisations such as boxplots and time-series plots show noticeable changes in communication behavior under attack conditions, especially during the DoS scenario. These results highlight the negative impact of cyberattacks on smart grid communication and show the usefulness of simulation for testing communication resilience. This paper provides evidence of how cyberattacks can affect communication behavior in smart grids and offers a method that can support future studies on attack detection and communication resilience in critical infrastructure.

Keywords: Smart Grid Security, SmartGridSim, Cyberattacks, Network Performance Analysis, Denial of Service (DoS), False Data Injection (FDI)

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1

Introduction

1.1 Introduction

The electricity sector is currently undergoing a significant shift from traditional centralized systems to digitalized power grid system. Although traditional power grids mainly support one-way power flow, modern smart grids use digital communication technologies to enable two-way flows of data and energy that improve efficiency, reliability, and system monitoring (Rouhani & Su, 2026). This digital transformation brings clear benefits of greater efficiency, integration of renewable sources, and the ability to monitor systems in real time (Hassine et al., 2025).

However, this dependencies on digital communication introduces new risks, such as disruption in information flow between substations, sensors, and control centers that can directly affect the reliability and stability of the electricity supply. These communication channels are critical and vulnerable to cyberattacks: Denial of Service (DoS) attacks can overwhelm network capacity with false traffic, while False Data Injection attacks (FDI) manipulate measurements in ways that mislead operators into taking incorrect actions (Y. Zhang et al., 2025).

Many existing security defenses struggle to keep up with the fast changing cyberattacks, despite years of development. False alarms and inconsistent detection performances remain common issues. Without practical thresholds, it is difficult for operators to distinguish regular network traffic from malicious behavior. This makes early attack detection more difficult.

For that reason, simulation-driven studies are important, especially for observing communication behavior and timing-related changes under controlled conditions. The present study addresses this gap by using SmartGridSim (SGSim). SGSim is an open source smart grid simulator designed for cybersecurity awareness (Holik et al., 2024). By applying both normal and attack scenarios in SGSim, network traffic is captured using Wireshark and later analyzed using Python tools. Metrics such as latency, jitter, packet loss, and inter-packet delay variation (IPDV) are tracked and compared, with a focus on identifying deviations from normal communication behavior.

Through this work, the study aims to provide insights into how cyberattacks influence smart grid performance, particularly in terms of communication timing and stability, and how measurable indicators can support system resilience. Although it does not introduce a new intrusion detection system, it strengthens the experimental foundation upon which future defensive strategies can be built.

1.2 Background and Related Work

Smart grids are replacing traditional power grid systems that are limited in many ways, for example, they mainly support one-way electricity delivery, lack real-time monitoring, and have limited ability to balance and distribute energy sources. The smart grid uses modern technology, digital communication, sensors, and automation that enable two-way flows of information and electricity, which is a major advantage over traditional power grids. However, this shift also exposes critical vulnerabilities in modern smart grids that did not exist in traditional systems (Rouhani & Su, 2026).

The IEC 61850 standard plays a central role in enabling digital substations. It defines communication protocols among intelligent electronic devices (IEDs) and supports services such as GOOSE messaging and sampled values. These features allow for real-time monitoring and protection. However, the same time-sensitive nature of IEC 61850 communication also makes it sensitive to even minor delays. Studies show that adversaries could exploit these delays to undermine protection mechanisms, particularly through FDI and DoS attacks (Elyamani et al., 2025).

Elyamani et al. (2025) identify observability as another major challenge and show that loss of observability during attacks directly reduces the accuracy of state estimation, compromising situational awareness. To mitigate this, some works propose observability indicators to detect when system visibility is impaired. Al-Qirim et al. (2025) also highlight the potential of cyber threat intelligence (CTI). For example, they combine knowledge graphs, digital twins, and hybrid machine learning to create a more realistic view of cyber risks in SCADA networks. This integration enables both detection and visualization of attack effects. More importantly, such approaches require proper validation before they can be adopted in practice.

False Data Injection (FDI) attacks are another serious threat to smart grid systems. This type of attack can inject or alter data so that operators make incorrect decisions, causing severe consequences. Z. Zhang et al. (2025) propose a conformal neural network model that uses probability-based conformity scores to separate false from normal data. Their work demonstrates accuracy above 97 percent and sensitivity above 93 percent. Similarly, research on adversarial attacks emphasizes the importance of robust anomaly detection systems. A framework combining GRU-based classifiers with Bayesian uncertainty quantification has been identified to achieve almost perfect detection of adversarial inputs (Efatinasab et al., 2025).

Simulation environments are widely recognized as effective tools to validate these ideas. Holik et al. (2024) developed SGSim to emulate IEC 61850-based communication and provide a testing ground for cybersecurity research and awareness. This simulator allows researchers to test attacks and defenses without risking real-world assets. In addition, Elbez et al. (2025) present insights from a realistic smart grid cybersecurity testbed that bridges the gap between simulation and physical systems, demonstrating how controlled experiments can uncover vulnerabilities and validate protection strategies.

In general, the literature indicates significant gaps remain in detecting anomalies and improving resilience. Many existing solutions struggle to adapt in real time, and false alarms remain a barrier. Simulation-driven research, therefore, continues to be necessary, particularly for establishing measurable and interpretable thresholds that can help operators detect and respond to attacks at an earlier stage (Hassine et al., 2025) and (Y. Zhang et al., 2025).

1.3 Problem Statement

Smart grids use modern communication technology and networks to improve efficiency, reliability, and automation. Although this digitalization relies heavily on real-time monitoring and control, it also creates new cybersecurity risks. Protocols such as IEC 61850, which are central to substation automation, have known vulnerabilities. Services such as GOOSE messaging can be exploited, where adversaries could delay or manipulate communication, threatening grid stability.

Two main types of attacks show these risks. Denial of Service (DoS) attacks can block or delay communication, reducing the reliability of the system. False data injection (FDI) attacks manipulate measurement data to mislead operators and compromise state estimation (Y. Zhang et al., 2025).

Although detection strategies exist, they often produce high false alarm rates and do not adapt to evolving attacks. Simulation-based methods show potential to fill this gap, but experimental evidence on how DoS and FDI affect communication performance and support measurable thresholds-based detection remains limited.

This project addresses the lack of simulation-based evidence on how Denial of Service (DoS) and False Data Injection (FDI) attacks degrade smart grid communication performance, the absence of practical thresholds to support early anomaly detection, and the limited analysis of system resilience.

1.4 Research Objectives

The primary objective of this project is to monitor and quantify the impact of cyberattacks on smart grid network performance using SGSim, with the aim of identifying measurable threshold for anomaly detection and system resilience.

To achieve this primary objective, The following six secondary objectives were identified. Firstly, the study examined the behavior of smart grid protocols IEC 61850, particularly GOOSE and sampled values (SV) under both normal and attack conditions. Secondly, SGSim was deployed and configured to replicate realistic substation communication. Third, baseline performance metrics, including latency, jitter, packet loss, and IPDV, were collected. Fourth, Denial of Service (DoS) and False Data Injection (FDI) attack scenarios were simulated within this controlled environment. Fifth, Wireshark and Python-based tools were used to capture and analyze traffic. Finally, thresholds for anomaly detection were examined based on the observed performance degradation.

These objectives align with the recent recommendations discussed in the literature review section. As Efatinasab et al. (2025) emphasize the importance of anomaly-based approaches to defend against adversarial attacks, while Z. Zhang et al. (2025) highlight the need for experimental validation to bridge the gap between theory and practice.

1.5 Research Methodology

The methodology used in this study investigates teh impact of cyberattacks on smart grid communication systems. It follows five interconnected phases: literature review, SGSim setup, baseline measurement, attack simulation, and data analysis. These phases aim to produce quantitative insights into performance degradation caused by DoS and FDI attacks.

The first phase of this project was a literature review to understand smart grid protocols such as IEC 61850, including GOOSE, and common cyber threats such as Denial of Service (DoS), False Data Injection (FDI) attacks, along with advanced defense mechanisms like cyber threat intelligence (CTI). Achaal et al. (2024) emphasize that identifying risks and protection strategies through proper assessment is fundamental to strengthening overall smart grid security. This part of the review also helps identify relevant metrics such latency, jitter, packet loss, and instantaneous packet delay variation (IPDV). In addition, Al-Qirim et al. (2025) highlight the importance of integrating CTI with digital twins and knowledge graphs to provide better visibility into attack behavior and system vulnerabilities. By critically analyzing these works, the project ensures alignment with current academic discussions and existing research directions.

The second phase of the methodology was to deploy the SGSim and configure the basic and extended topologies within a Mininet-based virtual network. SGSim has a built-in performance monitoring feature that was used to collect baseline data under normal communication conditions. This ensures that the simulation environment replicates the realistic communication patterns (Holik et al., 2024).

The third phase was the baseline measurement, and in this stage, the data was collected under normal operating conditions without any attacks. Key metrics such as latency, jitter, packet loss, and IPDV will be recorded. These measurements provide the reference against which later attack scenarios were compared.

The attack simulations were performed in the fourth phase of methodology using SGSim's native scripts. In this stage, DoS and FDI attacks were launched within the environment, targeting communication links that transmit GOOSE messages. During these simulations, Wireshark and SGSim's monitoring tools were used to capture network traffic and performance metrics in real time, enabling detailed observation of performance degradation.

The fifth and final phase focused on the collected data that were analyzed and visualized using Python tools such as Pandas and Matplotlib. This helps to clearly show how the network performance changes during different cyberattacks. The goal was to identify clear thresholds in the data where normal behavior transitions into unusual patterns, which can help detect problems early and improve the resilience of the system. Y. Zhang et al. (2025) emphasize the similar approaches where they validated in studies on FDI detection and adversarial robustness,confirming that thresholds can improve the reliability of anomaly detection

This methodology was chosen because it allows for systematic testing of cyber-threats without affecting real infrastructure. Simulation provides a safe and controlled space to change settings and see how the system reacts. SGSim was designed for smart grid cybersecurity and helps to test how attacks affect communication.

1.6 Scope and Limits

This project focuses on evaluating how Denial-Of-Service (DoS) and false data injection (FDI) attacks affect performance in IEC 61850-based communication. The study uses SmartGridSim to emulate IEC 61850 communication in a substation environment, applying both baseline and attack scenarios. Performance metrics such as latency, jitter, packet loss throughput, and inter-packet delay variation (IPDV) are measured to assess the impact of attacks.

The project does not aim to design or implement a new intrusion detection system. Instead, it provides experimental evidence that can support early detection by examining measurable thresholds. Broader topics such as cryptographic protocol design, large-scale deployment in real power grids, or hardware-based solutions are beyond the scope of the study. These boundaries help the project remain focused on simulation-driven experimentation and practical insights into smart grid cybersecurity.

1.7 Document Structure

This report is organized into six chapters. Chapter 1 introduces the research topic, background, and problem statement, research objectives, methodology, and the project's scope.

Chapter 2 reviews the relevant literature on smart grid communication, cyberattacks, simulation testbeds, and performance metrics used to analyse network behavior.

Chapter 3 explains the research methodology and experimental setup. It describes the simulation environment, attack scenarios, data collection process, and evaluation approach.

Chapter 4 presents the implementation and experimental setup in more detail. It explains the SmartGridSim and Mininet environments, the network topology, the attack model, and the data extraction process.

Chapter 5 presents the results and analysis. It compares baseline, DoS and FDI scenarios using packet loss, latency, jitter, IPDV, threshold analysis, boxplots, and time-series plots. Chapter 6 concludes the report by summarising the main findings, explaining how the research objectives were met, outlining the research contribution, and suggesting future work.

2

Literature Review

This literature review examines recent studies on how communication in modern smart grid systems behaves under normal operation and under cyberattack conditions. The reviewed studies focus on communication timing, cyberattack effects, and the use of simulation testbeds to study these changes. These areas are directly related to this present study because the project examines how timing behavior changes under baseline, DoS, and FDI conditions. The reviewed work also provides the academic background for the methodological choices explained later in the report.

Current grid communication operates as a digital layer that interacts closely with physical processes, making timing consistency important for stable operation. Holik et al. (2024) show this clearly in their experiments with SGSim, where IEC-104, GOOSE, and SV traffic are generated under controlled conditions. Their results demonstrate that these protocols rely on regular message timing, and even modest changes in delay or jitter can influence how devices coordinate. Docquier et al. (2023) points in the same direction, listing latency, jitter, packet loss and inter-packet delay variation as key indicators of communication performance because they directly affect the stability of substation control networks. Achaal et al. (2024) also notes that as substations become more digital and interconnected, they inherit more timing-sensitive attack surfaces. This makes communication behavior worth monitoring.

Timing changes are also discussed in work focusing on attack scenarios. Oinonen and Morsi (2024) analyses substation automation systems under DoS, reply, and manipulated-measurement attacks and reports that these attacks leave measurable traces in message intervals and signal timing before more obvious problems appear. Research on grid stability supports the same idea. Efatinasab et al. (2025) shows that small adversarial or abnormal inputs can influence downstream predictions before operators notice anything unusual, suggesting that early deviations may reveal issues or incidents caused by communication problems or cyber threats rather than by physical faults.

Since testing such scenarios directly on live systems is expensive and unsafe, many researchers rely on simulation or emulation. Holik et al. (2024) describes how SGSim provides a reproducible way to study IEC-104, GOOSE, and SV under normal loads and attack conditions. Elbez et al. (2025) also discusses the benefits of cyber physical testbeds and shows how controlled environments support analysis of system resilience. These studies support the use of simulation to record normal timing patterns and compare them with timing behavior during cyber incidents.

2.1 Communication Foundations

A significant trend identified across the literature is the shift in communication within the smart grid from a simple data pathway to a digital backbone that directly supports protection, monitoring, and control. Several authors describe how this shift means that devices across the grid now depend on steady, predictable message flows to coordinate their actions. Holik et al. (2024) illustrates this clearly in their SGSim study. They show how IEC-104, GOOSE, and Sampled Values (SV) messages together provide a near real-time picture of substation activity, and how these protocols operate with a degree of timing regularity that forms part of the wider control process. This differs from older protection systems, where communication links were considered secondary and not tightly connected to operational decisions. For the present study, this timing regularity is important because later chapters examine whether DoS and FDI cause visible changes in communication performance.

Recent research also shows how this development fits into wider industry practice. Docquier et al. (2023) notes that many contemporary substations rely on consistent timing, particularly in areas that handle protection systems and supervisory operations. Their review shows that small timing shifts, such as extra delays, jitter, or uneven packet spacing can alter how certain protection routines interpret the same event. Achaal et al. (2024) expands on this by noting that increased digitization has created stronger dependencies among communication paths. For that reason, irregular timing in one part of the substation can affect several devices at the same time, making it more challenging to treat communication behavior as separate from the physical process it supports.

Several studies also examine how the physical and logical layout of a digital substation affects communication behavior. Research on layered and virtualized architecture shows that bringing different protocol families together can affect how each layer handles traffic. Although SGSim does not model physical grid equipment, Holik et al. (2024) notes that the communication topology is maintained in the simulation, allowing messages to be traced through switches, gateways, and other devices. Elbez et al. (2025) reported similar observations from their cyber-physical testbed. In their case, the arrangement of the substation network shaped the way timing disturbances traveled across the system and influenced how operators interpreted what they saw. These observations help frame the following sections, where protocol behavior, attack effects, and the performance indicators used in this study are examined in more detail.

2.2 Communication Protocols in Digital Substations

In digital substations, GOOSE, Sampled Values (SV), and IEC-104 are the three most important protocols commonly used for protection, measurement, and supervisory control. Each one behaves differently on the network. GOOSE carries fast, event-driven protection signals that must arrive within a few milliseconds. SV messages deliver high-rate current and voltage samples and are very sensitive to small timing shifts. IEC-104 works at a slower pace and links the substation to the control center, and any change in its normal spacing pattern is usually noticeable when the network becomes busy. Because these protocols form most of the everyday traffic in the substation and each has a recognizable timing pattern, they provide a useful reference point for studying how DoS and FDI attacks alter normal message behavior. As the main communication standards used in digital substations, their timing characteristics form the foundation of the analysis in this project, where normal and under-attack inter-arrival patterns in SGSim.

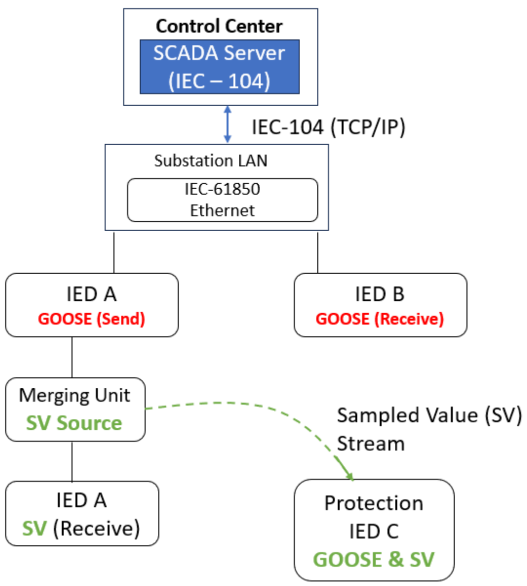

Figure 2.1: Substation communication architecture illustrating the flow of IEC-104, GOOSE, and Sampled Values (SV) messages.

Figure 2.1 helps show why the three protocols are useful for communication analysis in this study. IEC-104 links to control center with the substation LAN and supports supervisory communication. GOOSE is used for fast event-based protection messages between IEDs, while Sampled Values (SV) carries measurements data from the merging unit to receiving devices. Because these protocols serve different communication purpose and operate with different timing patterns, they provide a useful basis for observing how cyberattacks affect smart grid message behavior.

In the present study, the difference between these protocol roles is important because delay, jitter, or packet irregularity may appear differently depending on whether the traffic carries control, protection, measurement data.

2.3 Protocol Level Behavior

At the protocol level, IEC-104, GOOSE and Sampled values (SV) each shows their own timing behavior. These timing patterns help explain why communication delays or irregular spacing appear so clearly during cyberattacks. IEC-104 carries supervisory messages between the substation and the SCADA server. In SGSim, these packets follow a steady one-second reporting rhythm with occasional read requests, creating a predictable background pattern (Holik et al., 2024). Because the intervals are so regular, even small spacing changes can indicate network load or interference.

GOOSE traffic behaves differently. It is event-driven and designed for fast protection signaling. When an event occurs, the IED publishes repeated GOOSE frames at short and increasing intervals. Since GOOSE is sent as a layer-2 multicast with strict timing expectations, any irregular spacing or delay can affect protection actions (Docquier et al., 2023; Holik et al., 2024).

Sampled Values (SV) messages form the most time-sensitive stream. They carry high-frequency measurement samples at a fixed period rate and leave little tolerance for jitter. Earlier studies describe SV timing as highly reproducible, which makes deviations such as packet spacing changes or micro delays easier to detect (Docquier et al., 2023). In a digital substation, this is important because measurement traffic must remain stable for later processing and protection related functions.

Together, these protocols show three distinct timing signatures, these are slow periodic IEC - 104 cycles, fast GOOSE bursts, and dense SV samples. Small shifts in any of these patterns can signal performance degradation or an early attack. For that reason, these protocols behaviors form an important part of the analysis in the present study.

2.4 Cyberattack Impacts on Smart-Grid Communication

Cyberattacks often begin by disturbing communication patterns before any physical or visible malfunction appears. Several studies show that timing, message spacing and packet consistency can begin to change as soon as the network is under stress. Oinonen and Morsi (2024) shows this in their evaluation of replay, DoS, and manipulated measurement attacks. Each attack type produces noticeable changes in message timing, including shifts in arrival times, gaps between packets, and irregular sequence behavior. These changes often appear early, before a more obvious system response, which makes them useful indicators of abnormal communication behavior.

Efatinasab et al. (2025) report a similar pattern. Their results show that even small changes made to measurement values can influence later calculations, including signals that help relays decide how to respond. These adjustments often occur quietly, while the system still appears to operate normally, which allows an attacker to affect state estimation or control decisions without triggering an immediate alarm. Their findings suggest that communication-level distortions can reveal problems earlier than traditional monitoring alone would notice.

Broader review studies also support this pattern. Alomari et al. (2025) note that many cyber incidents first appear as minor performance issues such as queues forming on switches, slower SCADA updates, or packets disappearing rather than immediate system faults. These small timing changes reduce situational awareness and slow operator reactions, giving intruders more room to influence the system without relying on obvious attack activity.

Taken together, earlier research shows that cyberattacks leave a measurable signature in communication traffic while the system is still functioning. Small shifts in latency, jitter, packet spacing, and timestamp regularity can serve as early warning signs. This supports the approach in the current study, which compares baseline communication with DoS and FDI conditions in SGSim to observe when timing drift begins and how communication performance changes under attack.

2.5 Simulation Testbeds and Experimental Approaches

Most smart-grid research relies on simulation or small laboratory testbeds because real power systems cannot be used for experimentation. These controlled environments enable the study of timing behavior, load conditions, or cyberattacks without risking equipment. Holik et al. (2024) describes SGSim as one such setup. In their work, IEC-104, GOOSE, and SV messages are generated with stable timing, and the substation layout, protocol scripts, and device models are bundled inside a virtual machine. This arrangement allows the same experiment to be repeated many times, so any delay or jitter that appear can be linked to the test itself rather than outside influences.

One of the major benefits of simulation testbeds is that the same scenario can be run multiple times under the same conditions. In timing based research, even small setup differences can affect delay, jitter, or IPDV. For this study, the repeatability is important because the project compares baseline, DoS, and FDI runs using the same communication structure and performance metrics.

Other researchers use hybrid testbeds that mix virtual and physical components. Elbez et al. (2025) shows that when real hardware is added, timing irregularities become easier to spot, and operators can observe how attacks unfold in a setting closer to the field. Their findings suggest that testbeds serve not only a technical purpose but also help engineers understand attack behavior in practical terms.

Research papers also highlight the growing role of simulation across smart-grid studies. Docquier et al. (2023) describes the evaluation setups most often used and explains why emulation is well-suited for analysing communication behavior. Achaal et al. (2024) further notes that testbeds are increasingly utilized as substations adopt more digital components and face new security challenges.

Together, these works show why simulation and testbed research form the basis for analysing communication patterns in digital substations. They provide stable, repeatable conditions for measuring latency, jitter, packet loss and IPDV under both normal and attack scenarios. This is the same approach taken in the present study, where SGSim is used to record baseline performance and compare it against communication behavior during DoS and FDI attacks.

2.6 Performance Metrics

Recent studies show that communication performance is often the first place where problems appear, both under normal operating conditions and during cyberattacks. Docquier et al. (2023) note that metrics such as latency, jitter, packet loss, and IPDV are widely used because they show how reliably packets move through the network. Even when messages are still delivered, small changes in these values can point to problems on the communication link. These metrics are also practical to collect because they can be measured without interfering with power-system processes and are easy to quantify.

Findings from cybersecurity-focused studies support the same view. Oinonen and Morsi (2024) show that DoS, replay, and falsified measurement attacks can affect state estimation by increasing reporting delays and reducing successful communication. Efatinasab et al. (2025) also reports that abnormal network behavior can distort transmission before operators notice any visible problem. Both studies suggest that performance degradation often appears early in an attack, which makes timing-based metrics useful for early warning.

Several other papers also introduce additional indicators that build on basic performance measures. Docquier et al. (2023) highlights how performance measures such as delay, jitter, and packet consistency can be used to track changes in communication behavior under different operating conditions. Irfan et al. (2024) combine event-triggered state estimation with communication-based checks to detect signs of FDI behavior Zheng et al. (2025) look specifically at IEC-104 traffic and use an unsupervised method to identify behavior that deviates from normal patterns when labeled datasets are not available. Gutiérrez Mlot et al. (2024) assess a substation dataset and show that controlled testbed data can help detect subtle timing variation that would be difficult to capture in the field.

Together, these studies show that communication performance measures are a practical way to monitor how a smart grid network behaves over time. They are sensitive enough to reveal small deviations, but still simple enough to interpret across different scenarios. Nearly all of these approaches follow the same logic: first establish a reliable baseline, then compare it with communication behavior under attack. This is also the approach used in the present study, where baseline, DoS, and FDI scenarios are compared to identify early signs of communication degradation.

Findings from the cybersecurity-focused studies reinforce this view. Oinonen and Morsi (2024) shows that DoS, replay, and falsified measurement attacks affect state estimation by increasing reporting delays and reducing successful communication. Efatinasab et al. (2025) similarly reports that abnormal network behavior can distort transmission long before operators notice any visible problem. Both studies suggest that performance degradation often appears early in an attack, making timing-based metrics useful for early warning.

Several other papers also introduce additional indicators that build on basic performance measures. Docquier et al. (2023) highlights how performance measures such as delay, jitter, and packet consistency can be used to track changes in communication behavior under different operating conditions. Irfan et al. (2024) combines event-triggered state estimation with communication-based checks to spot signs of FDI behavior. While Zheng et al. (2025) looks at specifically IEC-104 traffic and uses an unsupervised method to flag behavior that deviates from standard patterns when labeled datasets are not available. Gutiérrez Mlot et al. (2024) assess a substation dataset and shows that controlled testbed data helps detect subtle timing variation, which would be challenging to capture in the field.

Together, these studies show that communication performance measures are a practical way to monitor how a smart-grid network behaves over time. They are sensitive enough to reveal minor deviations, yet straightforward to interpret across different scenarios. Nearly all methods rely on the same principle, such as establishing a reliable baseline, then looking for differences. This directly supports the approach used in this project, which compares baseline and under-attack behavior to identify early signs of communication degradation.

2.7 Gap Analysis

Although the reviewed studies offer valuable insights into how cyberattacks influence communication behavior, several gaps still remain that justify the present work. Across papers by Docquier et al. (2023), Efatinasab et al. (2025), Holik et al. (2024), and Oinonen and Morsi (2024), researchers consistently show that attacks can delay messages, increase jitter, or distort measurement values. However, most studies do not examine how these timing changes develop step by step in a controlled experiment where conditions remain identical before and during the attack.

Docquier et al. (2023) outlines timing budgets and measurement procedures, but they do not track how latency, jitter, packet loss, or IPDV evolve as an attack unfolds. Holik et al. (2024) recreated IEC-104, GOOSE, and SV traffic realistically in SGSim, yet they do not compare how these protocols drift from baseline once exposed to disruption. Oinonen and Morsi (2024) show that timing irregularities can signal replay, DoS, or FDI activity, but they do not identify which metrics shift first or how early that change becomes visible. Efatinasab et al. (2025) demonstrate how altered measurements affect relay behavior but do not link these effects to timing anomalies in the communication path.

A second gap is repeatability. Hybrid or hardware-assisted testbeds, for example Elbez et al. (2024) and Kabbara et al. (2025), can capture realistic timing effects from real switches and devices, but those same hardware details make exact repetition difficult. Slight differences in physical components or timing can alter results across runs, which limits the ability to run identical trials or isolate a single network effect. Using a software-only platform such as SGSim avoids this problem as every run can start from the same initial state, so any timing change observed is a direct outcome of the experiment rather than an uncontrolled hardware variation.

A third gap appears when comparing attack types. Many studies investigate DoS behavior alone, for example Elbez et al. (2024) and Sen et al. (2025), while others focus only on integrity-based manipulation such as FDI (Efatinasab et al., 2025; Luo et al., 2025; Zheng et al., 2025). It remains unclear how DoS and FDI differ in their timing effects when tested under the same conditions, as these studies use different setups and metrics. This leaves an important question unresolved: which metric, latency, jitter, packet loss, or IPDV, shows the earliest or strongest deviation for each attack type?

Overall, the literature recognizes the value of timing-based monitoring but lacks quantitative, repeatable measurements taken in a virtual testbed where baseline and attack scenarios are directly comparable. These gaps motivate the current study and help explain the methodological choices presented in Chapter 3.

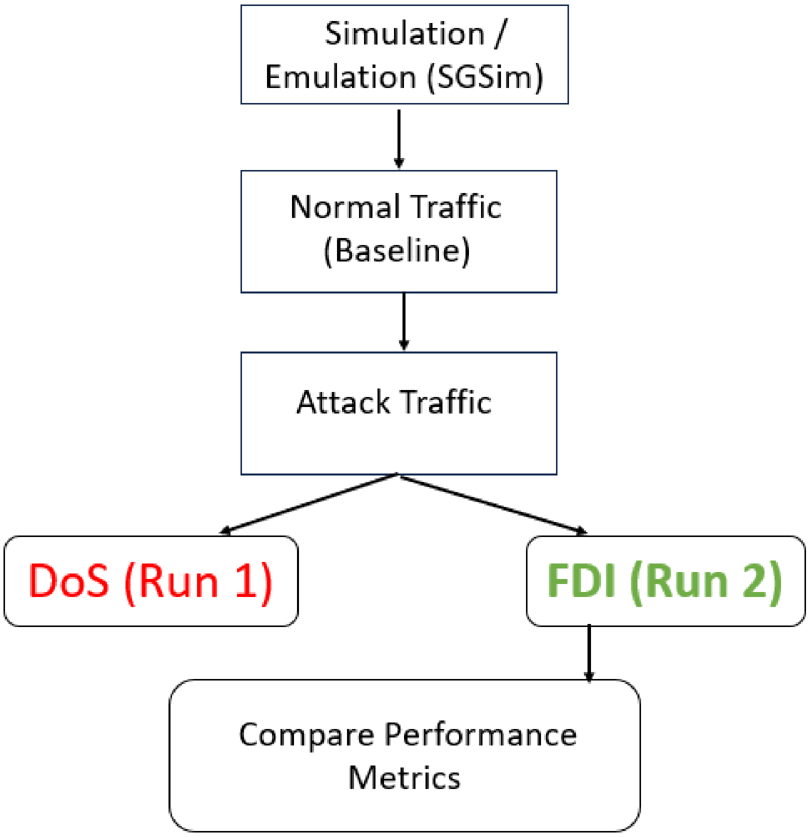

Figure 2.2: Workflow of baseline, DoS, and FDI runs in SGSim used for comparing communication performance metrics

Figure 2.2 shows the workflow used in the literature and followed in this project. The process starts with simulation in SGSim, then a baseline normal traffic condition is recorded before DoS and FDI attack are introduced. After that, the Dos and FDI runs are examined and the performance metrics are compared later from the collected data. The figure makes it clear that the comparison depends on having normal reference first and then checking how each attack changes communication behavior.

2.8 Summary of Literature Review

The studies discussed in this chapter make it clear that modern smart-grid communication relies heavily on precise timing. Cyberattacks can disrupt this behavior long before any visible malfunction occurs. Studies on DoS and FDI consistently report early irregularities in message timing, jitter, IPDV, and packet loss as well as changes in measurement values, suggesting that communication patterns often provide the first clues that something is wrong. Research on IEC-104, GOOSE, and SV further confirms that each protocol has its own timing signature and that changes in latency, jitter, packet loss, or IPDV can signal the onset of performance degradation. Studies using hybrid testbeds and Mininet-based platforms have improved understanding of substation communication and offered controlled environments for observing attack effects in practice.

Despite this progress, the literature still shows two apparent gaps: very few studies collect and compare communication metrics under identical baseline and attack conditions, and even fewer examine how timing behavior differs between DoS and FDI when both are tested in the same setup. These gaps define the motivation for the current study and shape the methodological choices introduced in Chapter 3.

3
Methodology

3.1 Methodology Overview

This project uses an experimental, simulation-based approach to study how cyberattacks affect communication inside a digital substation. Instead of relying on a physical smartgrid setup, which would involve costly equipment, safety rules, and access restrictions, the study was carried out inside a virtual laboratory. SmartGridSim (SGSim), which runs on Mininet, was used to build the main components of a substation, generating IEC 61850 traffic, and safely test different attack scenarios. Holik et al. (2024) shows that SGSim reproduces key features of substation communication well enough for research, and removes the obstacles that come with testing on real systems.

The basic idea of the method is straightforward, and follows this sequence: observe how the system behaves when communication is normal and then compare it with how it behaves when an attack is active. A stable baseline is recorded first, then two attack scenarios are introduced: Denial of Service (DoS) and False Data Injection (FDI). Each of these attacks affects the network differently. A DoS attack increases congestion and slows down traffic, while an FDI attack changes the original message values without necessarily affecting timing. (Oinonen & Morsi, 2024; Rouhani & Su, 2026).

There are four performance metrics that are used to measure how the system reacts. These metrics are latency, jitter, packet loss, and inter-packet delay variation (IPDV). Latency is the time it takes for a message to travel through the network. Jitter captures unpredictable

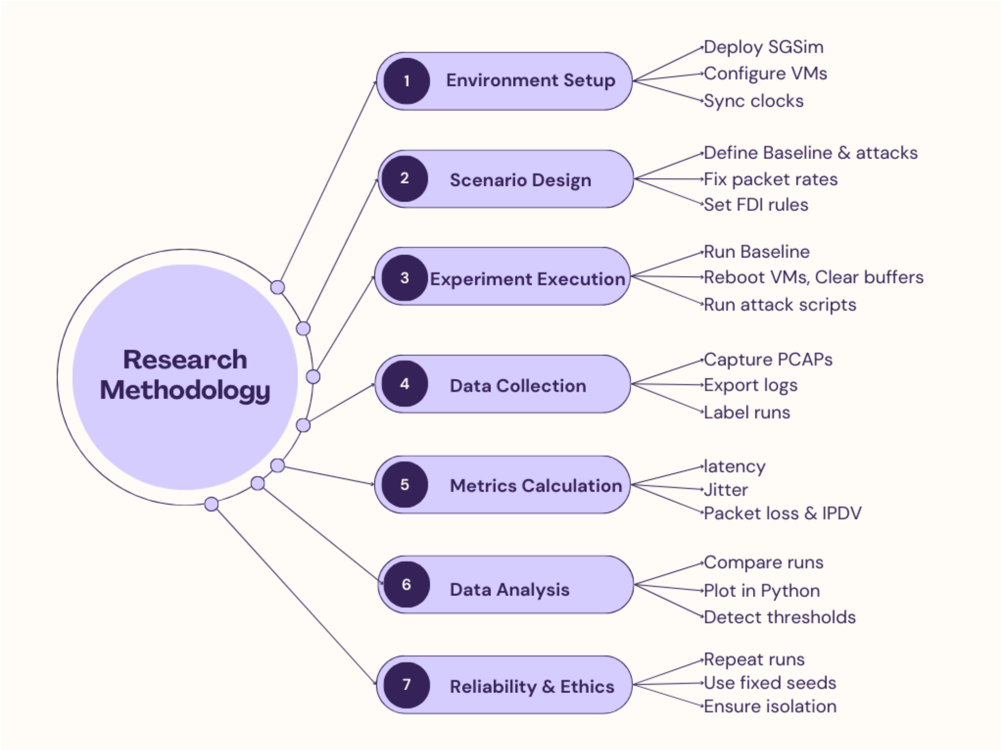

Figure 3.1: Methodology overview for baseline, DoS, and FDI experiments

changes in delay, which can affect time-sensitive communication in the substation network. Packet loss simply means messages that never arrive, and IPDV shows how evenly or unevenly packets are spaced during transmission. These four metrics are commonly used to assess the condition of communication links in critical infrastructure networks, and several studies recommend them as useful indicators of anomaly detection (Holik et al., 2024; Rouhani & Su, 2026). They also match recent anomaly detection work, which shows that unusual traffic behavior is often one of the first signs of an attack (Zheng et al., 2025).

The methodology follows seven steps. First, the SGSim environment is set up using VMware. Second, the attack scenarios are defined, including the baseline case and two attack types DoS and FDI. Third, the experiments are conducted one at a time, and each test begins with a system reset to avoid mixing traffic from the previous one. Fourth, data is collected from Wireshark captures and SGSim internal logs. Fifth, in this stage, the four performance metrics are calculated from the captured traffic. Sixth, in this stage, the normal and attack results are compared to identify any changes, patterns, or possible thresholds. Finally, in the seventh stage of the methodology, it ensures reliability and ethical considerations by ensuring that experiments are repeatable, safe, and fully isolated from the real system.

The goal of this methodology is to understand how cyberattacks affect smart-grid communication and whether early signs of degradation can be observed before complete communication failure occurs. Running the experiments in a controlled virtual environment made it possible to repeat the same setup, maintain the same timing structure, and compare the same communication path under different conditions. The methodology therefore provides a practical way to study communication degradation in smart grid systems without exposing real infrastructure to attack.

3.2 Environment Setup

The smart grid simulation environment for this study is built entirely inside a virtual smart grid testbed using SmartGridSim (SGSim). SGSim runs on Mininet, allowing the main elements of a digital substation, GOOSE publishers. Sampled Values (SV), intelligent electronic devices (IEDs), and a simple control room process are to be created without physical hardware. This virtual setup removes the usual constraints found in real substations, such as safety regulations, limited access, and high operational costs. At the same time, it still closely follows the communication patterns defined in IEC 61850 to support performance-focused research. As stated by Holik et al. (2024), SGSim offers a realistic and practical platform for studying substation communication without interacting with a live system.

The purpose of this framework was to observe how the communication layer behaves during normal operation and then compare it with its behavior when cyberattacks interact with the network. In digital substations, GOOSE and SV traffic carry time-sensitive messages and measurement data that must be delivered within milliseconds. Previous studies show that timing irregularities often appear well before any visible malfunction, making a controlled, repeatable environment valuable for identifying warning signs (Docquier et al., 2023). The SGSim environment was therefore a suitable environment to examine latency, jitter, packet loss and inter-packet delay variation under the same communication setup.

To maintain consistent conditions across all experiments, SGSim was deployed on virtual machines configured with same settings for all scenarios. The virtual machines used the same CPU and memory settings. Clock alignment and stable capture conditions were important because the performance analysis in later chapters depends on accurate packet timing. Keeping the environment stable in this way reduced the chance that timing results would be affected by unrelated system behavior.

The network layout remains unchanged throughout the study. SGSim was configured to represent a straightforward publisher-subscriber model. One IED publishes GOOSE and SV traffic, and another IED receives the traffic. This setup reflects how real substation devices share protection and measurement data, but without any extra routing that could complicate results. Because the layout is direct, timing is stable under normal conditions, which makes it easier to notice when delays or irregularities appear during the attack scenarios.

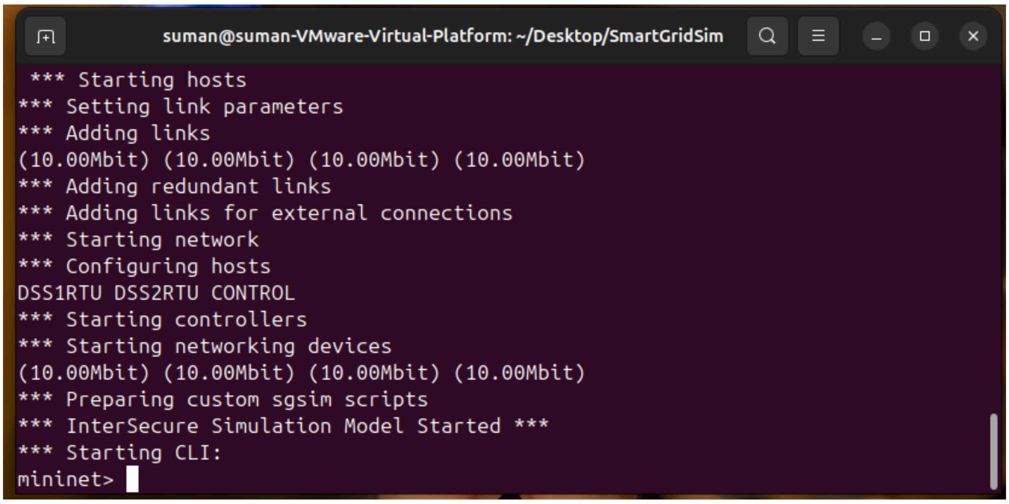

Figure 3.2: SmartGridSim environment setup and Mininet virtual network initialization

Figure 3.2 confirms that the Mininet virtual network was running successfully and the SmartGridSim environment was ready to start the experiments. The three scenarios Baseline (normal condition), DoS, FDI were prepared successfully before the experiments were executed. These three scenarios were prepared inside the same virtual environment. In the baseline scenario, SGSim generated normal communication traffic without attack activity. In the DoS scenario, additional traffic was introduced to overload the communication path and increase delay variation. In the FDI scenario, selected data values were modified while preserving the normal communication flow. Using the same environment for all three scenarios supported a fair comparison between normal operation and attack conditions. (Oinonen & Morsi, 2024).

There are certain preparations necessary before starting the actual attack scenario, for example, registering the current state of each machine, correcting the network configuration, and creating the VM correctly. These settings will prevent measurement errors. During each run, Wireshark captures all packets travelling across the network and SGSim's internal logs record arrival time, sequence numbers, and protocol details. These two data sources help to calculate latency, jitter, packet loss, and inter-packet delay variation accurately for each attack scenario.

Using the same network layout and timing configuration in all three scenarios supported a controlled and repeatable experiment. The environment made it easier to focus on how cyberattacks affect time-sensitive traffic and to observe early signs of performance degradation in smart grid communication.

3.3 Experimental Setup and Procedure

This experiment follows a clear sequence of steps so that each scenario runs in the same way, and it can also be repeated, and compared reliably. Every run begins with a reset of the virtual machine to ensure the next test is not affected because the reset of the virtual machine helps remove temporary files, network buffers, and leftover packets. This clean start is important in timing-based smart grid studies, as it maintains the stability of initial conditions across all four runs (Docquier et al., 2023).

Once the environment is reset, the baseline scenario is executed. SGSim generates continuous GOOSE and Sampled Values(SV) traffic at a fixed interval, creating a log for normal communication behavior. A brief verification process checks that sequence numbers follow the expected order and timestamps remain consistent across the devices. It is crucial to establish this baseline because it helps to detect changes in timing and message delivery once attacks are conducted (Oinonen & Morsi, 2024).

After completing and recording the baseline scenario, the DoS and FDI attack scenarios are executed. In the Denial of Service attack scenario, flooding traffic is generated toward the Control Center IP address using a Mininet ping flood command. This ping flood command created additional traffic on the communication path and allows the experiment to observe how availability-related disruption affects timing metrics such as latency, jitter, and IPDV. The same command and timing structure are used for each DoS run so that the results can be compared consistently. (Rouhani & Su, 2026; Y. Zhang et al., 2025).

Next, in the False Data Injection (FDI) attack scenario, a Python script is used to simulate false measurement values while keeping the normal communication flow active. This FDI scenario represents a data-integrity attack rather than a traffic flooding attack. The purpose is to compare whether the FDI scenario changes timing behavior differently from the DoS scenario. Full payload-level validation is outside the scope of this project (Zheng et al., 2025).

To keep a record of the fair comparison, each scenario is run for a certain period of time. A short pause is added between each run so that buffers can empty, and the system can stabilize before the next test begins. If any unexpected behavior occurs, such as CPU spikes, timestamp changes, or random errors unrelated to the attack, the scenario is repeated. All repetitions are documented to maintain transparency and support the reliability of the analysis. Throughout each run, Wireshark records all network packets, while SGSim logs internal events such as message types, arrival times, and performance counters. All capture files are saved in PCAP format and labeled clearly with the scenario name and timestamp (Holik et al., 2024).

After completing all scenarios, the captured data is prepared for metric calculation. The PCAP and log files are processed to extract latency, jitter, packet loss, and inter-packet delay variation (IPDV). In the next phase, the results from the baseline, DoS, and FDI scenarios are compared to identify early signs of performance changes or communication degradation. Because every test follows the same structure and conditions, the results provide a clear, repeatable basis for evaluating how cyberattacks affect smart grid communication (Rouhani & Su, 2026).

3.4 Attack Scenarios and Implementation

SGSim creates controlled attack scenarios to assess how cyberattacks affect the communication layer of a digital substation. These three scenarios focus on the most common threats to IEC-61850 communication. These threats are loss of availability and loss of data integrity. Each attack is applied individually, under real-time monitoring, in stable and consistent conditions, so that the communication behavior can be observed clearly. This approach allows the study to observe how the system responds to different types of disruption and how well the communication layer withstands its two most significant vulnerabilities.

3.4.1 Denial-of-Service (DoS) Attack

In the DoS scenario, the aim is to see what happens when the communication path becomes overloaded. A Mininet ping flood sends a large number of packets toward the Control Center at a fast rate. The same command was used each time, keeping each DoS consistent. As extra traffic builds, it competes with normal communication traffic for space on the channel. This can slow processing, create device queues, and sometimes cause packets to be dropped. Watching this behavior provides a clearer picture of how substation communication behaves under network pressure.

Delay and timing variation usually increase during this type of attack, although packet loss was not observed in the final results of this study. Earlier studies show that congestion tends to affect protective traffic long before any visible system failure (Oinonen & Morsi, 2024). By capturing the traffic during these stressful periods, the study can examine how latency, jitter, packet loss, and IPDV shift when the channel is pushed beyond its normal limits.

3.4.2 False Data Injection (FDI) Attack

The FDI attack scenario focuses on data accuracy rather than network load. In the FDI attack, an interceptor script captures outgoing simulated measurement values, generate false measurement values during the FDI scenario. The frame timing, structure, and sequence numbers are left untouched. So on the surface, the traffic looks completely normal.

Previous smart grid research showed FDI attacks as a subtle way of misleading monitoring tools or protection systems. They can introduce values that appear trustworthy while avoiding obvious metrics that are useful for identifying or delivery issues (Oinonen & Morsi, 2024). Because the timing remains stable, common network alarms may not react, which is why performance metrics are useful for identifying small irregularities.

In this part of the experiment, the study examines whether changing the measurement values produces noticeable differences in latency, jitter, packet loss, or IPDV. Even though an FDI attack does not slow down the network, small timing change or unusual packet gaps may still appear, giving an early indication that something is wrong.

Both attack scenarios run entirely within the virtual testbed and use fixed parameters, using consistent behavior across the attack period. After each run, the system is given a moment to stabilize before the next test. All captured traffic and logs are stored in a clear, organized format to facilitate later analysis. Both the DoS and FDI attack scenarios show how availability and integrity-based attacks can affect communication within a digital substation and highlight early signs of performance degradation relative to the baseline.

3.5 Data Collection and Metrics

Data collection begins once each attack scenario baseline, DoS, and FDI is running inside the SGSim environment. The aim of this stage is to gather the communication records needed to calculate the four performance metrics used throughout this study, and these metrics are latency, jitter, packet loss, and inter-packet delay variation (IPDV). These four metrics are commonly used in smart grid communication research because they help detect early timing changes, stability issues, and communication irregularities before any visible malfunction appears (Docquier et al., 2023; Elyamani et al., 2025). Using the same four metrics in all three scenarios also makes it possible to compare under normal and attack behavior under the same measurement structure.

During each test, Wireshark captures all network traffic exchanged between the publisher IED and the subscriber IED. Each capture is saved as PCAP file and labeled with the scenario name and timestamp. These PCAP files include frame arrival times, protocol types, source and destination addresses, and sequence numbers. At the same time, SGSim's internal logs corrupted their own timing information, such as message creation time, subscription time, and GOOSE, SV sequence counters. Combining Wireshark captures and SGSim logs helps align timestamps more accurately, which is important when analyzing small timing shifts in IEC-61850 communication. The same captured method is used in baseline, DoS and FDI runs so that the collected data can be compared directly later in the study.

After each scenario finishes, the PCAP files and log files are exported for processing. Extraction scripts using Tshark and Python are then used to extract the timing information needed for the four metrics: Latency, jitter, packet loss and IPDV.

These metrics are calculated as follows. Latency was measured using the

`tcp.analysis.ack_rtt` field extracted by `tshark`. This field represents TCP acknowledgment round-trip time, meaning the time between a TCP packet and its corresponding acknowledgment. Therefore, the latency values in this project should be understood as RTT-based timing measurements rather than synchronized one-way delay measurements. This approach was suitable for comparing the baseline, DoS, and FDI scenarios because all scenarios were captured and processed using the same method.

In the extracted CSV files, jitter and IPDV were calculated from the change between two consecutive latency values. The script first subtracts the previous latency value from the current latency value. This signed difference is stored as IPDV. If the difference is negative, the script converts it to a positive value for jitter. Therefore, IPDV shows the direction of the timing change, while jitter shows only the size of the change. (Oinonen & Morsi, 2024).

For example, if latency changes from 10 ms to 15 ms, the IPDV is +5 ms, and the jitter is 5 ms. If latency change from 15 ms to 10 ms, IPDV is -5 ms but the jitter is still 5 ms This example clearly matches the calculation in the metric extraction script.

$$\mathrm{IPDV}_i = L_i - L_{i-1}$$

$$\mathrm{Jitter}_i = |L_i - L_{i-1}|$$

Packet loss was approximated using the `tcp.analysis.retransmission` field extracted by `tshark`. In the extracted CSV files, packets marked as TCP retransmissions were given a `packet_loss` value of 1, while other packets were given a value of 0. Therefore, packet loss in this study should be understood as a retransmission-based packet loss indicator.

Each metric is calculated separately for the baseline, DoS and FDI scenarios. Since all tests use the same duration, network, layout, and timing configuration. The result can be compared directly. The processed data is stored in structured CSV files, making it easy to analyze them in the next phase of the study.

Using SGSim for data collection provides a clean and controlled environment in which timing behavior can be observed without the safety risks of working on real substation equipment. Any changes seen, such as delay spikes, jitter swings, missing frames, or irregular packet spacing, point directly to the effects of the DoS or FDI attack. These collected metrics therefore provide a clear basis for detecting early degradation in smart grid communication performance (Rouhani & Su, 2026).

3.6 Data Analysis and Evaluation

After collecting data from all three scenarios, the baseline, DoS and FDI, the next step is to examine the four performance metrics and observe how communication changes under attack. The aim is to compare the normal traffic with the attack traffic and find out when the timing starts to change. These small timing changes are usually visible before anything fails, which makes them helpful as early warning signs in smart grid communication (Docquier et al., 2023).

The analysis begins with the metric calculations from the CSV files produced in the attack phase. Each dataset contains latency, jitter, packet loss, and IPDV values for every scenario. The metrics are reviewed individually at first to understand how they behave in each scenario during normal operation and then compared against their behavior under stress. For example, in a DoS scenario, the communication link becomes overloaded, causing increased delay and stronger timing variation. On the other hand, the FDI scenario often keeps the timing stable but changes the underlying measurement values. When results are presented side by side, the differences between availability-based and integrity-based attacks become clearer (Oinonen & Morsi, 2024).

Next, the study examines how these changes unfold along the communication path. Rather than focusing solely on the final numbers. The evaluation examines where deviations begin and how they evolve as the attack continues. This step-by-step approach helps identify which parts of the message flow are most sensitive to disruption. Smart Grid researchers recommend this type of path-focused analysis because it can reveal early disturbance patterns even when packet loss is still low (Holik et al., 2024).

Visualization is an important part of this process. Using Python, line graphs and scatter plots are created to track how latency, jitter, packet spacing, and loss change over time. These plots make it easier to spot sudden jumps, gradual drifts, or unusual spikes. For example, increasing latency combined with irregular IPDV values can signal congestion from a DoS attack. While steady timing, but inconsistent measurement values may indicate an integrity problem, such FDI manipulation. These signs help determine whether the system is stable, beginning to degrade, or entering a risky state.

The analysis also checks whether the results repeat across multiple runs. If similar patterns appear again under the same conditions. It strengthens confidence in the findings and matches the emphasis on reproducibility seen in recent smart grid cybersecurity studies (Rouhani & Su, 2026).

Together, these steps form the basis for evaluating the next phase of the study. By comparing the baseline, DoS, and FDI results side by side, the study can highlight where the communication performance begins to slip and how each attack type affects the smart grid network. This provides clearer evidence of which timing changes may serve as indicators of anomalies during cyberattacks.

3.7 Reliability and Validity

Reliability in this study comes from running each scenario in a consistent, controlled manner so that the results can be trusted. The baseline, DoS, and FDI tests all use the same virtual machine, network layout, timing configuration, and duration. Before each run begins, SGSim is reset, the network buffers are cleared, and both VMs are rebooted. This prevents leftover activity from affecting the next set of measurements. If a run shows unusual behavior, such as a CPU spike or timestamp drift that is not caused by the attack, the scenario is repeated, and the outcome is recorded. Following this routine helps confirm that any patterns in the results are due to the attacks rather than background noise. This approach aligns with recent advice in smart grid simulation studies, which emphasizes the use of consistent test conditions and repeated trials to achieve trustworthy results (Holik et al., 2024).

Validity is considered from several perspectives. Internal validity is improved by changing only one factor between scenarios. In this project, the only variable that changes is the attack itself. All other conditions stay exactly the same across the baseline, DoS, and FDI runs. This makes it easier to identify whether the attack is responsible for changes in the performance metrics. External validity is enhanced by modelling timing-sensitive IEC-61850 communication that mirrors the real digital substation cannot behavior described in earlier research (Docquier et al., 2023). Although the simulation cannot capture every hardware-level detail, it reflects the key timing characteristics needed to study how messages move through the communication path. The four metrics, latency, jitter, packet loss, and IPDV are commonly used in laboratory tests and real-world networks as indicators of communication performance (Elyamani et al., 2025; Rouhani & Su, 2026).

There are few limitations found in the SGSim for example, it can not replicate the exact switching delays or hard-ware- specific behavior found in physical substations. Small differences, such as micro jitter or slight processing delays, may appear when compared to real equipment. The attack scripts are used in this study is simplified for safety, and a real attacker may use more advanced or concealed methods that behave differently. Both virtual machines also run on the same host, which may introduce minor timing offsets. These factors are recognized so that the results are interpreted carefully rather than treated as exact real-world values.

All experiments are performed inside a fully isolated virtual environment, which confirms the authenticity of the experiment. NO real substation, company network, or critical infrastructure is connected to the testbed, and the attack scripts cannot run outside this lab environment. The study does not use any personal or sensitive data, it focuses on communication in smart grid cybersecurity research, which highlights the need for safe, transparent, and responsible use of attack simulations (Rouhani & Su, 2026).

3.8 Chapter Summary

This chapter explains how this study is carried out in a virtual smart grid environment using SmartGridSim. It began with an overview of the approach and why a virtual testbed is a practical choice for looking at timing-sensitive communication. The simulation setup shows how SGSim recreates the key parts of a digital substation and why keeping the system settings consistent matters for getting fair and stable measurements.

This chapter also walks through how each experiment is run. All scenarios, baseline, DoS, and FDI follow the same setup and timing so they can be compared directly. Before each run, the SGSim environment is reset to clear out any leftover activity. DoS attacks introduce flooding traffic to test availability, while FDI attacks simulate false measurement values to test what happens when data integrity is targeted. During every scenario, Wireshark captures the network traffic, and SGSim logs the timing details needed for the analysis.

Next, the chapter outlines the four communication metrics used in the study, which are latency, jitter, packet loss, and IPDV. These are commonly used in smart grid research and help reveal early signs that the system is slowing down or becoming unstable. The chapter also described how the PCAP files and logs are processed to calculate these metrics for each scenario. Finally, reliability, validity, and authenticity are discussed. Running each scenario in the same controlled way supports reliability. Validity comes from modeling realistic IEC-61850 communication patterns and using well-known performance indicators.

Overall, this chapter presents a complete and repeatable methodology. These steps form the foundation for the next chapter, where the results from the baseline, DoS, and FDI scenarios are presented and examined to understand how each attack affects the smart grid communication path.

4

Implementation and Experimental Setup

This chapter describes the step-by-step processes of how the SmartGridSim (SGSim) environment was implemented, how the experimental setup used to evaluate communication performance in a smart grid system. The experiments showed how cyberattacks affect network behavior under controlled conditions. In this study, it introduces three attack scenarios that are normal operation which is Baseline scenario, Second, Denial of Service (DoS), and third is False Data Injection (FDI). All these three scenarios were executed in a Mininet based virtual network. In Mininet, IEC 104 communication is available between the control center and digital substations. Each experiment followed the same execution procedure to keep the consistency that allows comparison between normal and attack conditions.

While Chapter 3 described the research design and data collection method, this chapter focuses on the practical implementation of the SGSim environment, experiment scripts, and attack scenarios. The experimental setup uses the control side interface to capture network traffic that flows between substations, Control Center and HMI. This setup helps to provide a consistent observation for measuring performance degradation. The captured data was used to measure key performance metrics: latency, jitter, packet loss and IPDV. Each scenario was repeated 12 times, giving 36 runs in total for three different scenarios to ensure reliability and reproducibility. This chapter follows the following sub sections: the experimental environment, network topology, experiment workflow, data collection process and implementation of baseline and attack scenarios and finally data processing for the analysis in the next chapter.

4.1 Experimental Environment

The experiments were conducted inside an Ubuntu-based virtual machine running on VMware. The SmartGridSim topology was deployed using Mininet to emulate the communication network between the Control Center and digital substations. The SmartGridSim and Mininet setup provided a fully virtual and isolated environment where baseline, DoS, and FDI scenarios could be tested under the same conditions. Using the same virtual setup for all runs helped keep the experiments consistent and reduced the chance that outside factors would affect the results.

Table 4.1 summarises the main technical configuration used to run the baseline, DoS, and FDI experiments.

Configuration item	**Description used in this project**
Simulation environment	SmartGridSim / SGSim running inside an Ubuntu-based virtual machine.
Virtual machine specification	Ubuntu virtual machine configured with 8 GB memory, 4 processors, 80 GB disk, and NAT network adapter in VMware Workstation.
Network emulator	Mininet was used to run the simulated smart grid network topology.
Main communication point observed	Traffic was captured from the Control Center side of the network.
Capture interface	`CONTROLSW-eth3`, connected to the Control Center communication path.
Capture tool	Wireshark / `tshark`. Packet captures were saved as PCAP files and later processed into CSV files.
Experiment scenarios	Baseline, Denial of Service (DoS), and False Data Injection (FDI).
Runs per scenario	12 runs were collected for each scenario.
Capture duration	180 seconds per run.
Timing structure	30 seconds pre-period, 120 seconds main observation or attack period, and 30 seconds post-period.
DoS attack method	Flooding traffic was generated toward the Control Center using `mininet h1 ping -f 1.1.10.10`.
FDI method	The `inject_fdi.py` script was used to simulate false measurement values during the FDI scenario.
FDI value range	The script generated simulated false measurement values in the range 40.0 to 60.0.
Metric extraction	Metrics were extracted from the PCAP files using `extract_metrics.sh` and `tshark`.
Extracted fields	Timestamp, source IP, destination IP, frame length, TCP ACK RTT, retransmission flag, latency, jitter, IPDV, and packet loss indicator.
Analysis tools	Python 3 with pandas and matplotlib.
Main analysis outputs	Statistical summary table, boxplots, time-series plots, and threshold exceedance comparison.

Table 4.1: Experimental setup and configuration for the SGSim experiments
Source: Author

The main SCADA system in the topology is the control center (CONTROL IP 1.1.10.10), which interacts with the digital substations through WAN routers and gateways. Each digital substation contains the real-time communication path between the Remote Terminal Units (RTUs) and intelligent electronic devices (IEDs). These devices exchange measurement and control information within the internal substation network and forward relevant data to the control center. The environment uses the IEC 104 as the primary protocol for data exchange between the control center and the substations. Once the Mininet starts and the SGSim communication command is executed, communication is established between the control center and both substations before packet capture begins. Establishing communication before packet capture was important because the metrics in this study were based on real-time communication rather than idle network traffic.

All three experimental scenarios were carried out in the same virtual environment. Each run started from a clean state so that leftover traffic from an earlier run would not affect the next one. The notes from the practical execution show that the virtual machine was started fresh, the SGSim environment was activated, and the Mininet topology was launched before traffic capture began. A short stabilization period was then allowed before the main observation window. Keeping the same execution sequence for baseline, DoS, and FDI made the comparison more reliable.

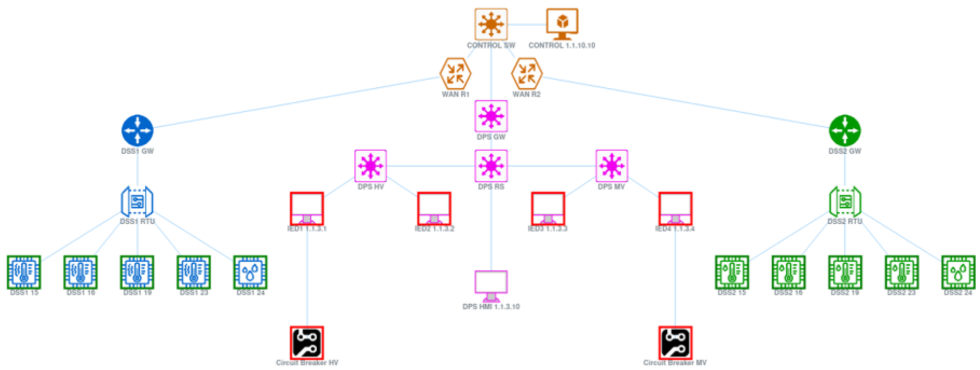

Figure 4.1: Network topology of the SmartGridSim testbed used in the experimental evaluation

Figure 4.1 shows the network topology used in the experimental evaluation. The topology illustrates how the SCADA control center is connected to the two digital substations through the communication network. The control side includes the CONTROL node, the CONTROLSW control switch, and the WAN routers that carry traffic between the control center and the substation. On the substation side, each site includes a gateway, a remote terminal units (RTU), and internal network segments connected to multiple IEDs and other substatioin devices. The network topology reflects the communication path used for monitoring, control, and data exchange during the experiments.

Within the network topology, the control switch plays an important role because it carries the traffic exchanged between the control center and the substations. The practical notes also show that traffic was monitored from the control side where all relevant communication passed through a single observation point. Monitoring traffic from the control side made it possible to analyze the network behavior from the same position in every scenario. As a result, the calculated latency, jitter, packet loss, and IPDV valued were based on a consistent communication path across all runs.

The topology was selected because it was detailed enough and from the actual experiment to represent control center to substation communication. This topology shows the real time communication between the control and substations. The exact same topology allowed the structured test applied across all scenarios and it supports a fair comparison between normal operation and attack conditions. The stable experimental structure gave the study a reliable foundation for the data collection and analysis presented in the following sections.

4.2 Network Topology

The simulated network represents a simple version of smart grid communication model that consists of control, network and digital substations layers. Figure 4.2 shows the SmartGrid-Sim topology used in the experiments. At the control layer the CONTROL node represents the SCADA system. The CONTROL node is connected to the rest of the network through the control switch CONTROLSW. The WAN routers R1 and R2 maintain communication between the control center and two substations.

Each digital substation contains a gateway, a remote terminal units (RTU), and internal network segments that connect multiple IEDs and other devices. In the topology the gateways are DSS1GW and DSS2GW, while the RTUs are DSS1RTU and DSS2RTU. The internal substations side includes several IEDs that simulate measurement, monitoring, and control functions inside the substations. The RTUs collect information from the IEDs and exchange data with control center through the communication network. The control center and the substations communicate using IEC 104 over TCP/IP, and the communication remains active throughout all experimental runs.

The topology also reflects the communication path that was analyzed during the experiments. Traffic generated by the substations passes through the gateway and WAN routers before reaching the control side. For that reason, the control side provides a useful observation point for monitoring the communication behavior across the full path between the substations and the SCADA system. Figure 4.2 also shows the node and link structure reported by Mininet, which was used to confirm that the topology was deployed correctly before the experiments started.

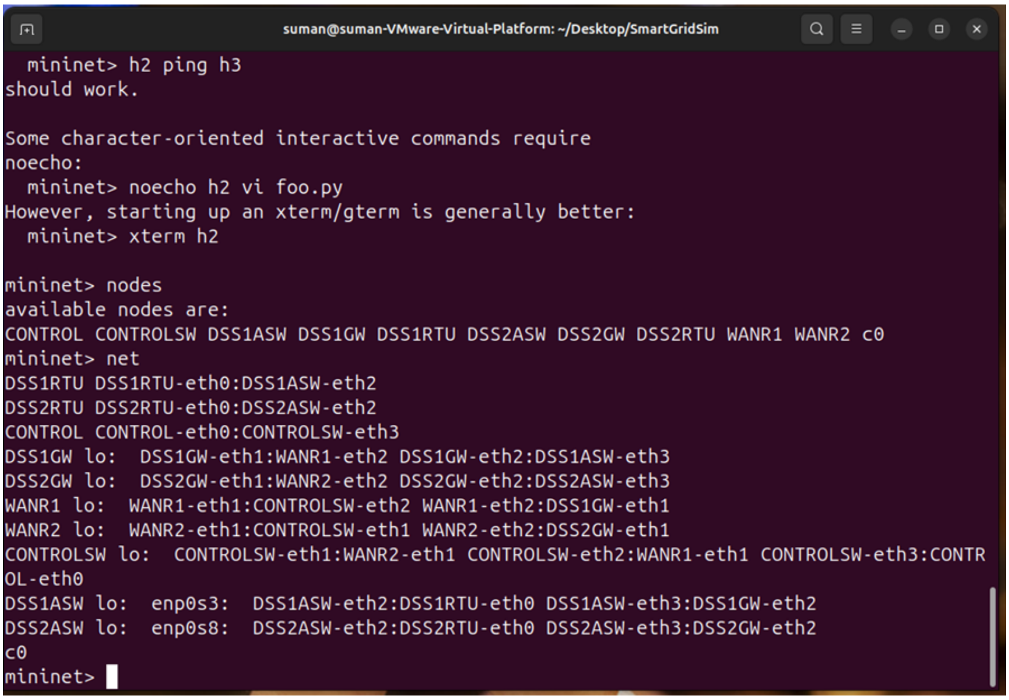

Figure 4.2: SmartGridSim network topology used in the experiments

The control side interface CONTROLSW -eth3 was used to capture all network traffic. The CONTROLSW-eth3 interface connects the control center to the network and provides visibility of incoming and outgoing traffic exchanged with the substations. Monitoring traffic at CONTROLSW-eth3 allowed the study to observe the same communication path in baseline, DoS, and FDI scenarios. As a result, the performance analysis was based on a consistent observation point across all runs. The SmartGridSim topology provided the communication structure used to evaluate latency, jitter, packet loss, and IPDV under baseline (normal condition), DoS, FDI attack conditions.

4.3 Experiment Execution Workflow

All attack scenarios including baseline condition, followed a fixed and repeatable workflow to ensure consistency throughout the experiment. After each run is completed, the next run starts with fresh virtual machine state. The StartDSSTopology.sh scripts activate the SGSim environment and the Mininet topology. Once the network is fully initialized, IEC 104 communication started with 4 different windows in the SGSim environment that shows the communication between the Control and substations. This ensures that everything is set properly to start capturing traffic.

The control interface (CONTOLSW-eth3) used to capture data packet. Each run follows 180 seconds of fixed timing. The first 30 seconds are to stabilize the system that allows us to reach normal operating conditions. The next 120 seconds used to observe either attack activity occurs or normal communication behavior. The final 30 seconds captured system behavior after the attack period or normal operation.

Each scenario ran for 12 times, giving 36 runs in total across the three attack scenarios. Outputs from all experiments were stored in a structured directory inside SGSim workspace. The main directories include pcaps/, csv/, and scripts/. The pcaps/ directory has three sub directories: baseline/, dos/, and fdi/. Each run generates a PCAP file and the name for each file format such as: `control_run_01.pcap`

Bash and python scripts were used to automate the attack period and all scripts located in the scripts/ directory. The main scripts are as follows: `run_baseline.sh` `run_dos.sh` `run_fdi.sh` `capture_control.sh`

The FDI attack was implemented using the python script saved as `inject_fdi.py` There are three open terminals, one terminal for running Mininet, another terminal for running the script and the third terminal to view the live communication in the SGSim. Following commands were used to execute runs:

```
cd /Desktop/SmartGridSim/experiments/scripts$
sudo ./run_baseline.sh 01
sudo ./run_dos.sh 01
sudo ./run_fdi.sh 01
```

The next command Tshark was used to verify the capture the data and data record. Example for dos run below:

```
sudo tshark -r ~/Desktop/SmartGridSim/experiments/pcaps/dos/
control_run_01.pcap -c 10}
```

The workflow above ensured that all experiments are repeatable and properly organized. It allows correct comparison between baseline and the attack scenarios.

4.4 Data Collection

Data was collected from the control side interface during every experimental run. Network traffic was captured in PCAP format so that the communication between the control center and the digital substations could be recorded in a consistent way across baseline, DoS, and FDI scenarios. Capturing traffic from the control side made it possible to observe the same communication path in all runs and provided a stable source of data for later metric calculation.

The captured data included packet timestamps, source and destination address, IEC 104 protocol information, frame details and TCP protocal and retransmission information. These packet-level records were later used to calculate latency, jitter, packet loss, and inter-packet delay variation (IPDV). Raw PCAP files were kept first so that the original network traffic remained unchanged before any processing or filtering was applied. Keeping the original captures was useful for later checking and analysis.

All captured PCAP files were stored in a structured way inside the SGSim experiment directories. The files were separated into three categories: baseline, DoS, and FDI. Each scenario was repeated 12 times , which produce 36 PCAP files in total. Each file follows the same naming structure. The only difference in the run number as `control_run_01.pcap`. This makes it easier to track and process the data later. The PCAP files were stored separately for baseline, DoS, and FDI so that each run could be tracked and processed correctly during later analysis. The organized file structure also reduced the risk of mixing data from different scenarios.

4.5 Attack Model & Implementation

Three experimental scenarios were used including the baseline, DoS, and FDI condition were implemented in this study. These scenarios were used to compare communication performance under normal operation and under two attack conditions DoS and FDI. The baseline scenario provided the normal state without attack activity. The DoS scenario was used to examine how communication timing changes when network traffic is increased. The FDI scenario was used to examine how communication behavior changes when transmitted values are manipulated without directly creating congestion.

4.5.1 Baseline Scenario

The baseline experiment shows normal system operation without any cyberattack. In the baseline phase, SmartGridSim runs under standard conditions, and IEC 104 communication between the control center and both substations occurs without disruption.

Each baseline run was set for 180 seconds. The first 30 seconds were used for stabilization. This was followed by 120 seconds of normal communication, and a final 30 seconds for post stabilization. All communication traffic was recorded in a PCAP file using an automated script while Mininet remained active in a separate terminal.

The captured PCAP packets were later used to extract data like timestamps, protocol details, and sequence information in order to calculate latency, jitter, packet loss, and IPDV. These baseline results were later used for comparison with the DoS and FDI attack scenarios.

The baseline scenario was executed using the following command:

```
sudo ./run_baseline.sh 01
```

4.5.2 Denial of Service (DoS) Attack Scenario

In the Denial of Service (DoS) attack experiment, network load was intentionally increased to create congestion in the communication channel. A Python script was used to send high rate traffic during the main observation window while IEC 104 communication between the control center and substations continued to run. The aim of the DoS scenario to observe how heavier traffic affected delay and timing behavior across the communication path.

The timing structure remained the same as the baseline scenario. First 30 seconds for stabilization, 120 seconds attack period, and last 30 seconds for the post stabilization. DoS traffic was captured on the CONTROLSW-eth3 interface and saved PCAP files for each run.

During the attack window, latency values became higher and timing fluctuations were more visible. Based on the extracted metrics, packets were still delivered successfully and no packet loss was observed. The impact of the DoS scenario was therefore mainly on delay and timing behavior rather than complete communication failure.

The DoS scenario was executed using the following command:

```
sudo ./run_dos.sh 01
```

4.5.3 False Data Injection (FDI) Attack Scenario

The False Data Injection FDI experiment focused on changing or manipulating measurement data rather than disrupting communication. The FDI scenario followed the same timing structure as the baseline and DoS scenarios. During this attack scenario, a Python script modified selected values within the communication stream and modified them before they reached the control center.

The overall packet structure, sequence numbers, communication flow and timing behavior were not intentionally changed by the FDI attack. Communication between the control center and substations continued normally at the network level. For that reason, the FDI scenario was expected to affect data correctness more directly than delay or traffic load.

Since the attack did not increase traffic or create congestion, its impact was less immediate than in the DoS scenario. The main effect was on data correctness rather than delay or availability. Table 4.2 summarises how the FDI scenario was represented in this experiment.

The FDI scenario was mainly included as a data-integrity attack. The timing results show that FDI remained closer to baseline than DoS, which is expected because FDI does not necessarily create heavy traffic congestion. The packet-level analysis in this project focused on timing metrics rather than deep payload validation.

Configuration item	Description
FDI script	`inject_fdi.py`
Value range	Simulated false measurement values between 40.0 and 60.0.
Injection interval	One simulated value was generated every 1 second.
Purpose	To represent a data-integrity attack where incorrect measurement values are introduced during the FDI scenario.
Timing metrics analyzed	Latency, jitter, IPDV, and packet loss.
Main interpretation	The FDI scenario was analyzed mainly in terms of communication timing behavior, not full payload-level validation.

Table 4.2: FDI scenario configuration used in the experiment
Source: Author

The FDI scenario was mainly included as a data-integrity attack. The timing results show that FDI remained closer to baseline than DoS, which is expected because FDI does not necessarily create heavy traffic congestion. The packet-level analysis in this project focused on timing metrics rather than deep payload validation. The project datasets record the FDI scenario timing results, while the project records the FDI configuration and simulated value range. Therefore, the FDI results should be interpreted as communication-timing behavior during the FDI scenario. The FDI results should not be treated as complete payload-level verification of false data manipulation.

The FDI scenario was executed using the following command:

```
sudo ./run_fdi.sh 01
```

The data generated from these three attack scenarios was then processed for further analysis as described in the next section.

4.6 Data Extraction and Preparation

The capture data was processed in a structured manner. Each PCAP file was converted into CSV format, and Python scripts were used to extract and aggregate the metrics into three master CSV datasets corresponding to the Baseline, DoS, and FDI scenarios.

The conversion process from PCAP to CSV was automated using scripts stored in the experiment workspace.The extraction process used packet-level fields such as timestamps, source and destination address, frame length, latency related values, retransmission information, jitter, IPDV, and packet loss indicators. The script extract metrics.sh was used to read each PCAP file and write the selected values into CSV format. For example, the following command was used to extract metrics from PCAP files:

```
./extract_metrics.sh \
../pcaps/baseline/control_run_01.pcap \
../csv/baseline/baseline_metrics.csv
```

After the individual PCAP files were processed, the extracted outputs were grouped into three master CSV datasets for baseline, DoS, and FDI. These scripts were used to process:

```
./process_baseline.sh \
../process_dos.sh\
../process_fdi.sh
```

Each master dataset contains Latency, Jitter, Packet Loss and IPDV measurements collected from the corresponding scenario. The processed files were then transferred to the analysis directory where statistical summaries and visual plots were generated using Python tools such as Pandas and Matplotlib. Keeping the extraction stage separate from the later analysis stage helped preserve original capture files and reduced the risk of changing the raw experimental data.

The extraction workflow was also checked carefully during the project. Each step was documented carefully on the practical notes. In the experiment process there was an issue on the jitter and IPDV result due to a small error in the script. The script was corrected carefully and tested on copied VM before making changes in the actual experiment. After the correction and testing, the PCAP files were processed again so that the final CSV dataset reflected the updated extraction logic. Reprocessing the PCAP files helped confirm the final analysis was based on checked and regenerated outputs.

The data extraction and preparation stage converted raw PCAP files into structured metric dataset that could be compared across baseline, DoS, and FDI scenarios. The prepared datasets were then used in the next chapter for statistical comparison, visualization, and interpretation of communication behavior under normal and attack conditions.

4.7 Summary

This chapter presented the implementation and experimental setup used in the study. The chapter explained how the SmartGridSim and Mininet environment was prepared, how the network topology was organized, and how the experiment workflow was carried out under controlled condition. The control side capture point at CONTROLSW-eth3 was used in all runs so that communication behavior could be observed from the same switch port across baseline, DoS, and FDI scenarios.

The chapter also described how the three scenarios were executed using the same timing structure and run the experiment. Each run followed a 180 second window with stabilization, main observation, and post run periods. Baseline, DoS, and FDI traffic were captured in PCAP format and stored separately in an organized directory structure. This made the runs easier to manage and supported an authentic comparison between normal and attack conditions.

The final part of the chapter explained how the captured PCAP files were converted into CSV datasets for later analysis. Python scripts were used to extract latency, jitter, packet loss, and IPDV values from the traffic captures and group them into master CSV datasets for each scenario. The extraction workflow was also checked and corrected during the project so that the final datasets were based on reviewed outputs. These processed datasets provide the basis for the performance analysis presented in the next chapter.

5

Results/Analysis

This chapter presents the communication performance observed during the SmartGridSim experiments carried out under three conditions: baseline operation, Denial of Service (DoS), and False Data Injection (FDI). The findings are based on 36 experimental runs, including 12 repetitions per scenario. For each run, traffic was captured at CONTROLSW-eth3 interface and stored in PCAP format. The captured data was processed to extract latency, jitter, IPDV, and packet loss values. The results were then combined into master CSV files for comparison. These metrics were used to evaluate communication behavior and timing performance in the smart grid environment.

The previous chapters explained the study background, the research problem, the experimental setup and the data collection process. This chapter builds on that foundation by presenting the results found and interpreting them in relation to the research questions and hypotheses introduced earlier in the report. The aim is to compare behavior under normal and attack conditions.

To ensure a structured and research driven analysis, the extracted results plots are interpreted to answer the following hypotheses:

- **Hypothesis 1 (H1):** Under baseline conditions, communication between the Control center and Digital substations remains stable, with low delay and limited timing variation.
- **Hypothesis 2 (H2):** A Denial of Service (DoS) attack increases network congestion, and leads to higher latency, greater jitter and unstable timing behavior.

- **Hypothesis 3 (H3):** A False Data Injection (FDI) attack has a smaller effect on timing-related performance than a DoS attack. It mainly affects the integrity of transmitted data.
- **Hypothesis 4 (H4):** The four core performance metrics: latency, jitter, packet loss and IPDV help distinguish between normal operation and attack scenarios.

This chapter presents the quantitative results obtained from the processing stage, including statistical summaries, distribution plots and time series analysis across the three experimental conditions. The purpose is to establish a clear comparison between baseline, DoS and FDI conditions before discussing each metric in more detail in the following sections.

5.1 Overall Statistical Summary

Before presenting individual metric analysis, this section provides an overall comparison of communication performance under baseline, DoS and FDI conditions. The overall results show that the baseline condition is the most stable, the DoS scenario causes the strongest timing disturbance while the FDI scenario remains closer to baseline behavior. The comparison uses the processed CSV data from all three scenarios. Each scenario was repeated 12 times, captures a total of 36 PCAP. All scenarios followed the same capture and processing steps.

Scenario	Runs	PCAP Files	Duration per Run (s)	Total Duration (s)	Packet Records
Baseline	12	12	180	2160	8286
DoS	12	12	180	2160	9073
FDI	12	12	180	2160	9029
Total	36	36	180	6480	26388

Table 5.1: Scenario and Dataset Summary Used in Chapter 5 Analysis
Source: Author

Table 5.1 shows the size of the dataset used for the Chapter 5 analysis. The study used three scenarios, baseline, DoS, and FDI, and each one was repeated 12 times under the same setup. This gave a total of 36 PCAP files. Each run lasted 180 seconds, so each scenario produced 2160 seconds of capture traffic and the whole experiment produced 6480 seconds in total. The baseline scenario produced 8286 packet records after processing completion while the DoS scenario produced 9073 and the FDI scenario produced 9029. This gives a total of 26388 processed records for the final comparison. The small difference record count between the three scenarios is important because is shows that all scenarios changed the communication behavior. The dataset is large enough to support repeated comparison across the three conditions and to reduce the risk that one short run or one isolated burst would affect the final result. It also support the repeatability of the experiment, because the same number runs and same duration were used in all scenarios. For this reason, Table 5.1

helps show that the later comparison in latency, jitter, packet loss, and IPDV are based on a balanced set of captures rather than a single run or uneven dataset.

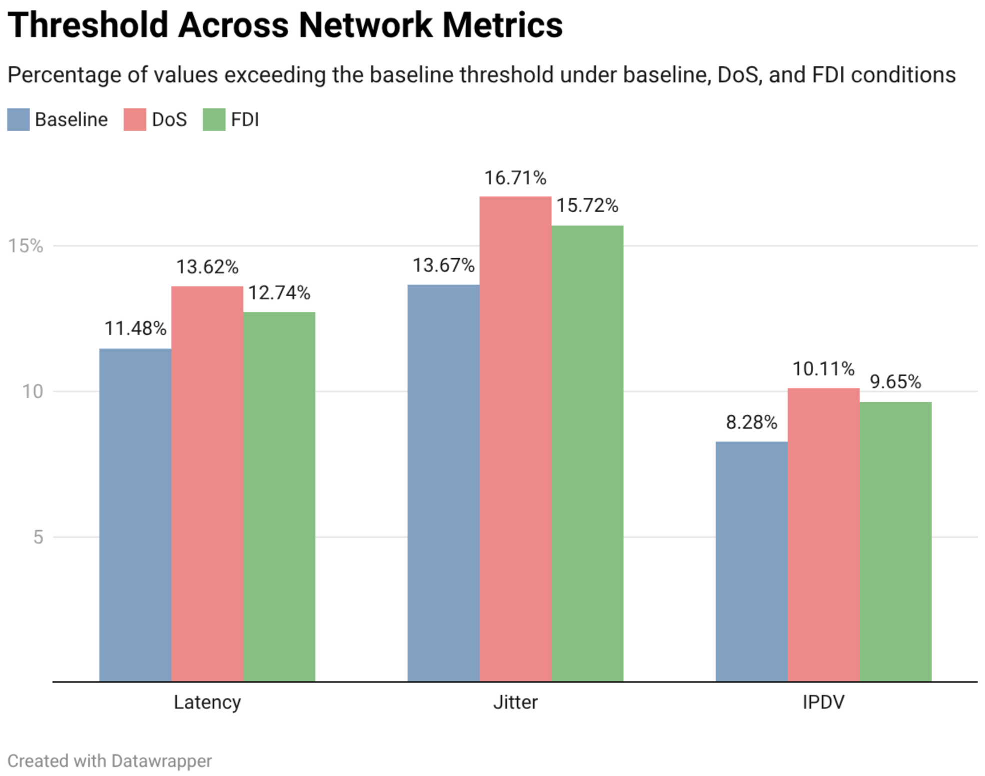

Figure 5.1: Percentage of values exceeding the baseline threshold across network metrics

To add statistical support to the threshold comparison in Figure 5.1, a two-proportion z-test was applied to the number of values above the baseline 2σ threshold in each scenario. This two-proportion z-test was used because the comparison focuses on the proportion of metric values that exceeded the baseline threshold under baseline, DoS, and FDI conditions.

Metric	**Baseline $> 2\sigma$**	**DoS $> 2\sigma$**	**FDI $> 2\sigma$**
Latency	951/8286 (11.48%)	1236/9073 (13.62%)	1150/9029 (12.74%)
Jitter	1133/8286 (13.67%)	1516/9073 (16.71%)	1419/9029 (15.72%)
IPDV	686/8286 (8.28%)	917/9073 (10.11%)	871/9029 (9.65%)

Metric	**Base vs DoS**	**Base vs FDI**	**DoS vs FDI**
Latency	2.09×10^{-5}	0.0112	0.0780
Jitter	2.78×10^{-8}	1.52×10^{-4}	0.0700
IPDV	3.25×10^{-5}	0.00167	0.299

Table 5.2: Two-proportion z-test results for values above the baseline 2σ threshold
Source: Author

Table 5.2 shows that the DoS and FDI scenarios produced higher proportions of values above the baseline 2σ threshold than the baseline scenario across latency, jitter, and IPDV.

A p-value below 0.05 was treated as statistically significant in this comparison. The baseline to DoS comparisons were statistically significant for all three metrics, and the baseline to FDI comparisons were also statistically significant at the 0.05 level. This means that the packet-level threshold test separated the attack scenarios from baseline more clearly than it separated DoS and FDI.

The z-test was applied to the merged packet-level CSV files. Therefore, it should be read as additional support for the threshold comparison, not as full run-level validation. A stronger method would be to calculate summary statistics for each of the 12 runs per scenario and compare the scenarios at run-level.

The baseline mean plus two standard deviations threshold was used as an exploratory comparison method. It helped compare how often each scenario moved outside the normal baseline timing range. However, some baseline values exceeded the threshold, as shown by the baseline exceedance percentages in Figure 5.1 and Table 5.2. This means that the threshold may produce false positives if it is used directly as an anomaly detection rule. For this reason, the threshold should not be treated as a deployable detection method. Future work could test percentile-based thresholds or robust statistical methods such as the median and median absolute deviation (MAD).

Figure 5.1 shows how often values exceed the baseline threshold for latency, jitter, and IPDV. The threshold was calculated using the baseline mean and standard deviation for each metric. The same threshold was then used to check the baseline, DoS, and FDI results.

In the baseline case, some of the values also exceed the threshold. Latency exceeds the thresholds by 11.48%, jitter by 13.67% and IPDV by 8.28%. This does not indicate an attack as the baseline represents normal condition. Baseline values can still exceed the threshold because it is calculated from statistical variation rather than the maximum observed baseline value. Therefore, the figure is used to compare the timing variation between the scenarios.

In the DoS case, jitter has the highest exceedance at 16.71%, followed by latency at 13.62% and IPDV at 10.11%. This shows that the DoS condition has the strongest effect on timing across all three metrics. The FDI case shows slightly lower than DoS, with 15.72% for jitter, 12.74% for latency, and IPDV at 9.65%.

From this comparison, DoS appears to have the strongest effect on communication timing. FDI also shows some variation, but its values stay closer to baseline than DoS. This fits the experimental design, where DoS increases the traffic load and FDI mainly changes the transmitted data values. The next sections look at packet loss, latency, jitter, and IPDV in more detail.

Metric	Baseline Mean	Baseline Std	2σ Threshold	Baseline > 2σ (%)
Latency	0.004978	0.012836	0.030649	11.48%
Jitter	0.008162	0.015640	0.039442	13.67%
IPDV	-0.000023	0.017642	0.035260	8.28%

Metric	DoS Mean	FDI Mean	DoS > 2σ (%)	FDI > 2σ (%)
Latency	0.005851	0.005531	13.62%	12.74%
Jitter	0.009783	0.009341	16.71%	15.72%
IPDV	-0.000029	-0.000030	10.11%	9.65%

Table 5.3: Statistical Summary and Baseline 2σ Threshold Comparison
Source: Author

Table 5.2 gives the exact values behind Figure 5.1 and shows how each scenario changed from the baseline timing pattern. For latency, the baseline mean was 0.004978 seconds and the 2σ threshold was 0.030649 seconds. Values above this threshold appeared in 11.48% of baseline records, 13.62% of DoS records, and 12.74% of FDI records. The latency value shows that the DoS scenarios caused the highest latency change, while the FDI scenario remained slightly lower and closer to baseline. The same pattern can be seen more clearly in jitter. The baseline mean was 0.008162 seconds and the threshold was 0.039442 seconds. Jitter went above this threshold in 13.67% of baseline records, 16.71% of DoS records, and 15.72% of FDI records.

The jitter value shows that it had the clearest difference between the three scenarios. For IPDV, the percentages were lower overall, but the same order remained: 8.28% in baseline, 10.11% in DoS, and 9.65% in FDI. This table 5.2 supports the same result shown in Figure 5.1: baseline remained the most stable condition, DoS caused the strongest timing disturbance, and FDI also changed timing behavior but stayed slightly closer to normal communication. The table is also useful because it shows the actual threshold values, not only the percentages, which makes the comparison clearer and supports the discussion in the next sections.

Table 5.3 shows the mean and standard deviation values for latency, jitter, and IPDV in the baseline, DoS and FDI scenarios. For latency, the baseline mean was 0.00498 seconds. The latency values show that both attack scenarios increased the average delay, and the DoS scenario gave the highest value. The same pattern appears in jitter. The baseline mean was 0.00816 seconds, while DoS increased it to 0.00978 seconds and FDI to 0.00934 seconds.

The jitter values also show that DoS had the strongest effect, while FDI stayed slightly closer to the baseline condition. For IPDV, the mean values in all three scenarios stayed very close to zero. However, the standard deviation values were 0.01764 in baseline, 0.01934 in DoS, and 0.01889 for FDI. The IPDV values show that the average value did not change much, but the spread became wider during the attack scenarios.

Metric	Scenario	Mean (s)	Std Dev (s)
Latency	Baseline	0.00498	0.01284
Latency	DoS	0.00585	0.01376
Latency	FDI	0.00553	0.01342
Jitter	Baseline	0.00816	0.01564
Jitter	DoS	0.00978	0.01669
Jitter	FDI	0.00934	0.01642
IPDV	Baseline	≈ 0	0.01764
IPDV	DoS	≈ 0	0.01934
IPDV	FDI	≈ 0	0.01889

Table 5.4: Statistical Summary of Mean and Standard Deviation Across Network Metrics
Source: Author

Overall, The table 5.3 follow the same pattern seen in the figure and threshold results. Baseline remained the most stable condition, DoS gave the strongest timing change, and FDI also changed the traffic but stayed slightly closer to normal communication.

5.2 Packet Loss Results

Packet loss was checked using the retransmission-based packet loss indicator in the processed CSV files. The packet loss check was done after the PCAP files were processed into CSV format. The goal was to check whether any retransmission-based indicators appeared.

In all three scenarios, packet loss remained at 0.000%. It confirms that no packet loss was observed in the baseline, DoS, or FDI datasets. All transmitted frames were successfully received at the control side interface CONTROLSW-eth3 during the 36 experimental runs.

This results confirms that the tested attacks did not cause complete packet delivery failure in the simulation environment. Even during the DoS scenario, packets were delivered in full. However, this does not mean that DoS scenario had no effect. The later timing results show changes in latency, jitter, and IPDV even though packet delivery remained stable.

In the FDI scenario, the packet loss is also expected. The FDI attack mainly changes transmitted date values rather than blocking communication. It affects mainly the data integrity. Because the FDI attack does not block communication, packet delivery can remain normal while the information being transmitted is affected.

Therefore, packet loss was not a strong indicator for distinguishing the scenarios in the study. The more useful difference appear in the timing-based metrics, especially latency, jitter, and IPDV. These metrics are later analyzed in the following section.

5.3 Latency Results

For each scenario, latency values were extracted from the processed dataset. The values were visualised using boxplots to compare distribution features and variability across baseline, DoS, and FDI conditions.

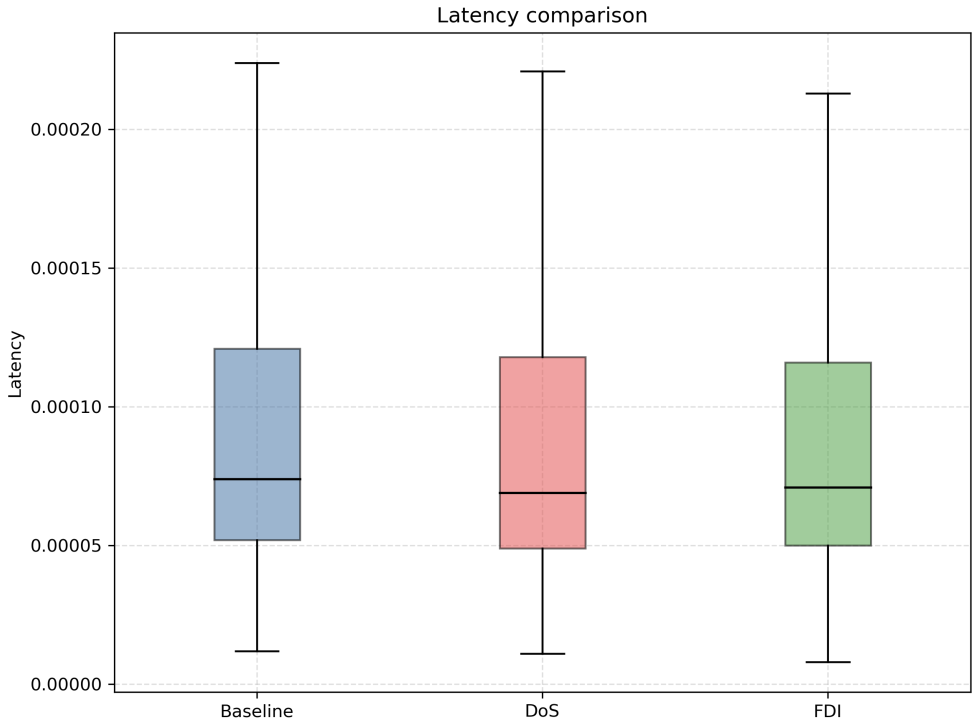

Figure 5.2: Latency comparison across Baseline, DoS, and FDI scenarios

Figure 5.2 shows that the median latency values remain close across the three scenarios baseline, DoS, and FDI, as seen in the similar median lines and box ranges. The similar median lines suggests that the middle range of latency does not change strongly between the three scenarios. The spread of the values shows how stable or unstable the timing behavior becomes under each condition.

In the baseline scenario, latency stays relatively stable. The mean latency is 0.004978 seconds, and most values remain within the lower range in the boxplot. This distribution is consistent with normal communication behavior, where packets travel through the network without noticeable delay changes.

However, the DoS scenario has the highest mean latency at 0.005851 seconds compared to 0.004978 seconds in the baseline case. This shows that DoS caused higher delays. The higher mean value increases the average delay and reduces timing stability.

The FDI scenario remains close to baseline, with a mean latency of 0.005531 seconds. Although there is small visible variation, the pattern remains less disturbed than in the DoS case. Therefore, the FDI attack had a smaller effect on communication timing than the DoS attack.

The latency results show that DoS had the clearest impact on delay behavior while FDI remained closer to normal operation.

5.4 Jitter Results

Jitter was measured by examining how the delay changed from one packet to the next. This helps show whether communication timing remains steady or becomes more uneven during the experiment.

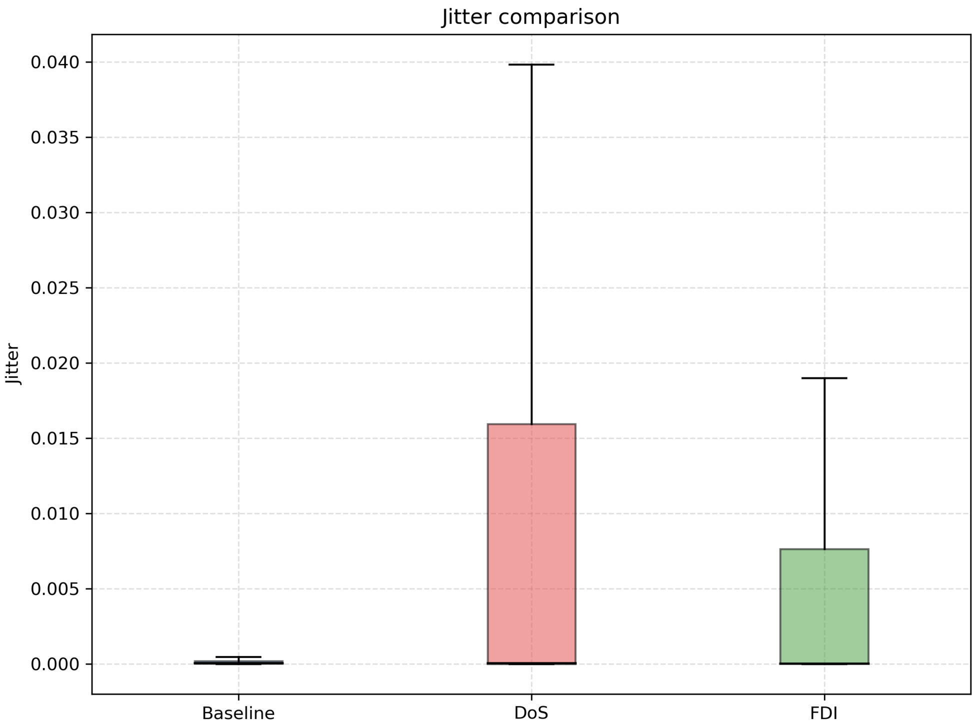

Figure 5.3: Jitter comparison across Baseline, DoS and FDI scenarios

Figure 5.3 shows the jitter distribution for baseline, DoS, and FDI scenarios. The median values are similar in all three cases, which can be seen from the median lines in the Figure 5.3 boxplots. However, the jitter values are spread differently. The baseline has the smallest spread, while the DoS scenario has the widest range.

In the baseline case, the mean jitter is 0.008162 seconds. Most values stay in the lower part of the boxplot, showing stable timing under normal conditions. The baseline box is very narrow compared with DoS and FDI. This confirms that packet delay changes were small during normal operation.

However, the DoS scenario has the highest mean jitter at 0.009783 seconds. It also has the widest upper spread, reaching close to 0.040 seconds in the boxplot. The larger box and the longer upper line show that jitter values were more widely distributed under DoS. This shows the strongest short-term timing variation under attack conditions.

The FDI scenario also shows more variation than the baseline case, with a mean jitter of 0.009341 seconds. Its box is wider than the baseline box, which shows that jitter increased under FDI as well. However, the upper spread reaches only about 0.019 seconds, which is much lower than in the DoS case. This shows that FDI affected timing variation, but not to the same level as DoS.

The jitter results show that DoS had the strongest effect on timing variation while FDI stayed closer to baseline behavior.

5.5 IPDV Results

Inter-packet delay variation (IPDV) was used to examine how the delay varied from one packet to the next. Jitter shows the size of the change. IPDV also shows whether delay increased or decreased. This makes it useful for understanding short-term timing changes in packet delivery.

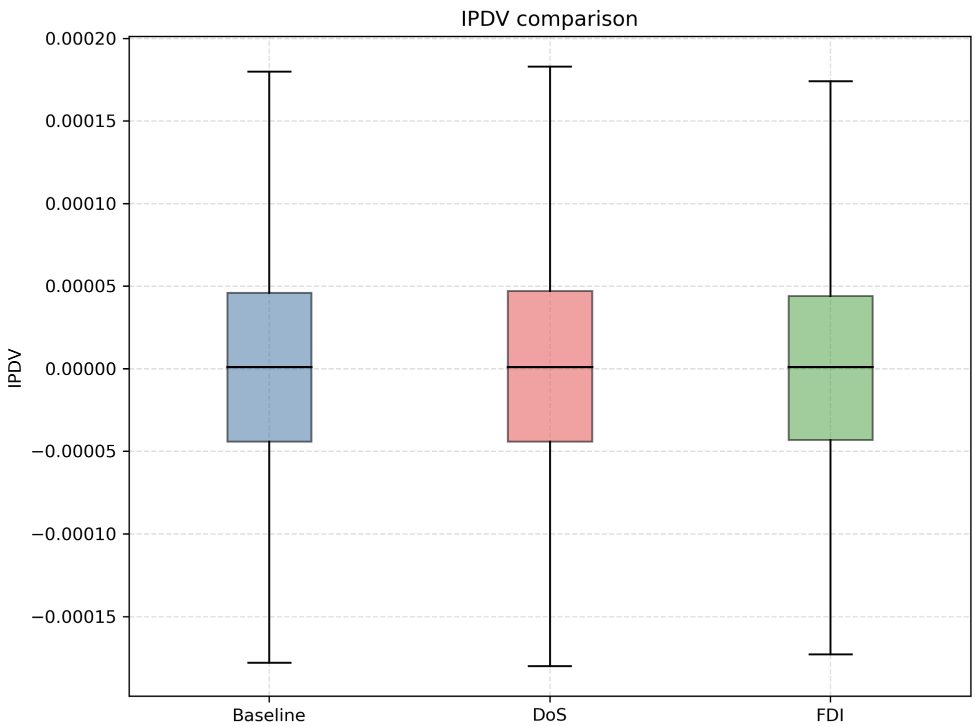

Figure 5.4: IPDV comparison across Baseline, DoS and FDI scenarios

Figure 5.4 shows the IPDV distribution across baseline, DoS, and FDI scenarios. The median lines stay close to zero in all three cases. The box ranges are also very similar. This is expected because IPDV is a signed value, so positive and negative changes can occur. From the Figure 5.4, the mean IPDV of baseline scenario is about -0.000023 seconds. The median line stays very close to zero, and the box remains narrow around the center. This shows that short-term delay changes stayed small during normal operation.

In the DoS scenario shows the mean IPDV is about -0.000029 seconds. This is still very close to zero, but the DoS case shows slightly more variation overall. The box and the spread line extend a little more, which suggests stronger short-term delay fluctuation during the attack. It matches the threshold result in Section 5.1, where DoS reached 10.11% IPDV exceedance.

By contrast, the FDI mean IPDV is about -0.000030 seconds. It is also close to zero and remains similar to the baseline pattern. The spread in FDI is slightly lower than the DoS case, although it still shows some variation. This is consistent with the threshold comparison, where FDI reached 9.65% exceedance, which is lower than DoS.

The IPDV results confirm that all three scenarios remain centred near zero. The DoS still has the strongest effect on the short-term timing variation. FDI also affects the timing pattern slightly, but less than DoS.

5.6 Time Series Analysis

Time-series plots were used to examine how latency, jitter, and IPDV changed during a 180-second run in each scenario. In the previous sections, the boxplots summarized the full distribution of values. The time-series plots make it easier to see how timing behavior changes during a run.

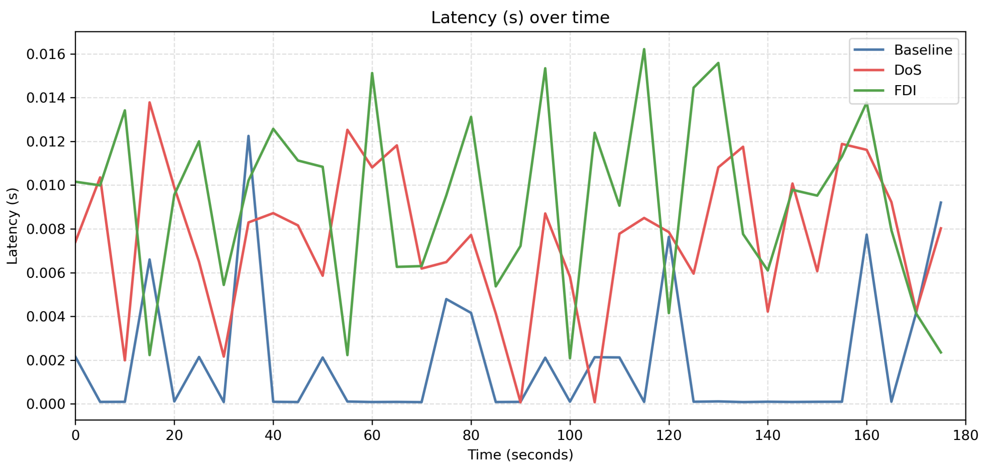

Figure 5.5: Latency variation over time across Baseline, DoS and FDI scenarios

Figure 5.5 shows the latency values over time. The x-axis covers the full 180 second run, while the y-axis show latency in seconds from about 0.000 to 0.016. The baseline run stays lower and more stable for most of the run. most of the values remain close to 0.000 to 0.002 seconds, although a few short peaks still appear. One clear example appears around 35 seconds, where the latency rises about 0.012 seconds.

In the DoS attack scenario, it is clearly seen that the DoS shows inconsistency with fluctuations in spikes over the full run period of window. Most DoS values remain around 0.006 to 0.012 seconds. Some peaks rise significantly to about 0.014 seconds. These peaks are mostly visible in the following time seconds of 20, 50, 60, 130 and 160. This confirms latency under DoS did not remain steady and changes over time than in the baseline run.

By contrast, the FDI run also shows noticeable fluctuation in latency. Several values rise above 0.015 to 0.016 seconds which is the highest point. This shows that both DoS and FDI scenarios introduce latency variation over time compared with the Baseline run.

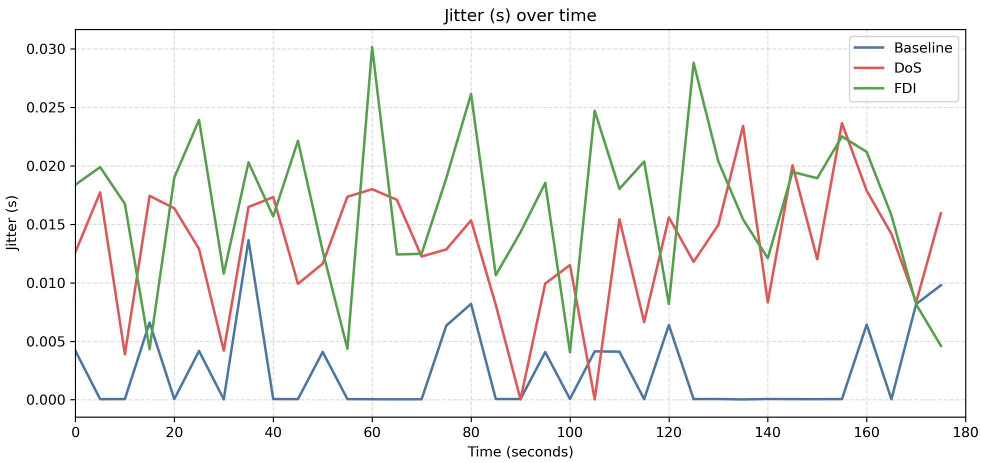

Figure 5.6: Jitter variation over time across Baseline, DoS and FDI scenarios

Figure 5.6 shows the jitter values over time. The x-axis covers The full 180-second attack period, while the y-axis shows jitter in seconds from about 0.000 to 0.030. In the baseline scenario, most values remain close to zero for most of the time window. It often stays very close to 0.000 seconds. There are few short rises visible but not lasted long. In around 35 seconds, a clear rise appear and reaches about 0.014 seconds.

From figure 5.6, it is clearly visible that the DoS run affects the jitter line across the run. Multiple DoS values stay around 0.010 to 0.018 seconds, and some peaks rise above 0.020 seconds. There are also clear rises seen around 55, 60, 135, 145, and 155 seconds. This shows that jitter under DoS changed more sharply from packet to packet than in the baseline run.

By contrast, the FDI attack scenario also shows noticeable jitter variation. Several values are visible rises above 0.020 seconds, and some peaks reach about 0.025 to 0.030 seconds.

This confirms that FDI also affects short-term timing changes during the run. However, DoS case still remains more unstable than FDI based on the results shown in the figure 5.6.

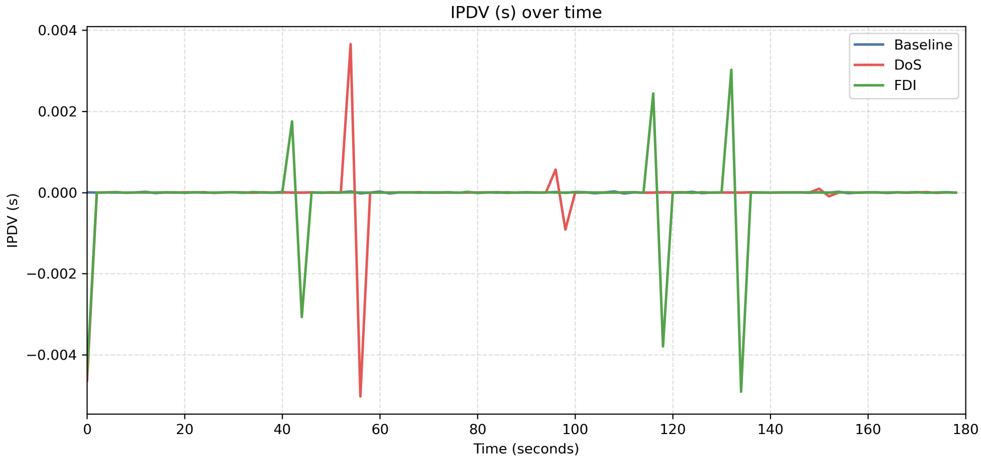

Figure 5.7: IPDV variation over time across Baseline, DoS and FDI scenarios

Figure 5.7 shows the IPDV behavior over time. The x-axis represents the full 180-second attack period, while the y-axis shows IPDV in seconds from about -0.005 to +0.004. In the baseline case, most values remain very close to zero except one larger negative value at the beginning. The line stays near the center for most of the run.

In the case of Denial of Service (DoS) scenario, the IPDV line shows sharper reverse changes around the middle of the run. One of the clearest changes occurred around 55 to 57 seconds, where the value rises to about +0.0037 seconds and then drops to about -0.0050 seconds. Smaller changes also appear later. This characteristic shows that packet to packet delay changed more abruptly under DoS, even though the average IPDV remained close to zero.

By contrast, the FDI scenario shows visible spikes in IPDV. There are clear visible changes around 42 to 45 seconds and again around 115 and 135 seconds. Some peaks rise to about +0.0025 to +0.0030 seconds, while some negative drops reach about -0.0040 to -0.0050 seconds. This shows that FDI also produces short-term timing fluctuation during the attack phase. However, the larger figure still shows FDI stays slightly below the DoS case.

5.7 Chapter Summary

This chapter examined communication performance under Baseline, DoS, and FDI conditions using statistical summaries, boxplot-based distribution comparisons, threshold analysis and time-series analysis. No packet loss was observed in any of the three scenarios, which shows the packet delivery remained stable during the experiments. The clearer differences appeared in the timing-based metrics, especially latency, jitter, and IPDV.

The results show that the DoS scenario has the strongest effect on timing behavior. It produced higher latency, wider jitter variation, and stronger short-term variation in IPDV than the baseline case. The FDI scenario also showed some timing variation, but it remained closer to baseline than the DoS case.

The findings show that DoS had a stronger effect on communication timing while FDI was designed to affect transmitted data values and data integrity, although this project evaluated it mainly through timing behavior.

6 Conclusion

6.1 Introduction

This chapter presents the main findings of the research and explains what was learned from the full experimental process. The study examined how cyberattacks affected communication performance in a simulated smart grid environment built with SmartGridSim and Mininet. Three scenarios were tested: baseline operation, Denial-of-Service (DoS), and False Data Injection (FDI).

The project was carried out in a controlled virtual environment. Each scenario followed the same run structure, timing window, and capture process. A fresh virtual machine state was used before each run so that no leftover traffic or earlier system activity affected next run. This controlled setup helped keep the comparison fair and made it easier to judge whether a change in timing was caused by the test condition itself.

The final results showed a clear difference between the two attack types: DoS and FDI. The DoS scenario had the strongest effect on communication timing. It increased delay, reduced timing stability, and spread the variation in latency, jitter, and IPDV. The FDI scenarios also showed some timing variation, but it remained closer to baseline than to DoS in most comparisons. At the same time, packet loss stayed at 0.000% in all three scenarios. This means the attacks did not stop packet delivery in the tested environment, but they still affected communication timing stability. This section summarizes the main findings and links them to the research objectives.

6.2 Summary of Research

Chapter 1 introduced the research problem and explained why communication security in smart grids matters. Smart grid systems increasingly rely on digital communication between control centers and substations. Because of these dependencies, attacks that affect communication timing or transmitted values can reduce reliability and make monitoring more difficult. The chapter defined the aim of the project, which was to examine how cyberattacks affect communication performance in a controlled smart grid simulation. It also presented the research questions and hypotheses that guided the study.

Chapter 2 discussed key literature on smart grid communication, IEC 61850 traffic, cyberattacks, and timing-based monitoring. The literature showed that DoS and FDI attacks affect smart grid systems in different ways. DoS is usually linked to congestion and timing disruption, while FDI is more closely linked to data integrity. Studies by Holik et al., Docquier et al., Rouhani and Su, Hassine et al., and Zhang et al. highlighted the need for controlled experiments to study how cyberattacks affect communication systems.

Chapter 3 presented the research methodology and experimental design. The experiments used the SmartGridSim (SGSim) environment on Mininet to create a virtual environment for testing baseline, DoS, and FDI scenarios under controlled conditions. The same communication path, timing window, and capture method were used in all scenarios. Each run followed the same sequence: Start from a clean state, allow 30 seconds stabilization, execute the attack period for 120 seconds, and allow a short post-run stabilization for 30 seconds before stopping the capture. This structure was important because the project was based on timing-sensitive metrics. Chapter 4 explained the implementation and data processing workflow. Traffic was captured at the control side of the network, where the communication from the substations could be observed consistently in one place. Capturing traffic at the control side gave a stable capture point for comparing all three scenarios. The captured PCAP files were then processed using Tshark and Python scripts. The extracted values were stored in CSV files and used for statistical comparison, boxplots, thresholds analysis, and time-series analysis.

Chapter 5 interpreted the final results. Packet loss remained at 0.000% in all scenarios. The baseline condition stayed the most stable. The DoS condition showed the strongest timing disturbance across latency, jitter, and IPDV. The FDI condition also showed some timing variation, but its values remained closer to baseline than to DoS. The threshold comparison supported the same pattern. The time-series plots added details by showing when peaks and reversals happened during the 180-second runs.

6.3 Research Objectives

This research paper mainly focused on monitoring and quantifying the effects of cyberattacks on communication performance in a simulated SmartGridSim (SGSim) environment. The objective was achieved by building a repeatable test environment in SGSim and Mininet and by comparing communication behavior under baseline, DoS, and FDI conditions. The study did not rely on single run or one isolated result. Each scenario was repeated 12 times, which produced 36 PCAP captures in total. Repeating the scenarios under the same communication path and timing structure enabled comparisons across all three runs.

Measuring the communication performance using latency, jitter, packet loss, and IPDV was another objective of this project. The project successfully extracted these four metrics from the captured traffic and used them to compare the three scenarios. Packet loss was not as useful as other metrics because it remained 0.000% in all three scenarios. In contrast, the timing-based metrics showed clearer differences, especially in the DoS case. The timing-based metrics show that communication degradation can still be identified even when packet delivery remains intact.

A further objective was to use statistical analysis and visualization to make the difference between scenarios easier to observe. Statistical summaries, boxplots, threshold comparison, and time-series plots were used for this purpose. These methods showed that the DoS scenario caused the strongest timing disturbance, while the FDI scenario remained closer to baseline.

A final objective was to propose threshold-based metrics for anomaly detection using baseline metric behavior. This objective was addressed through the threshold comparison presented in Chapter 5. The threshold analysis showed how often values moved beyond the expected baseline range. The threshold comparison provided a practical way to compare normal and attack conditions and interpret communication degradation under cyberattack.

Within the planned scope, the project met the objectives by providing a controlled comparison between normal and attack conditions. It showed how communication metrics, threshold comparison, and visual analysis can be used to study smart grid behavior under cyberattack.

6.4 Research Contribution

This paper contributes to smart grid cybersecurity research in a few practical ways. One contribution is a controlled comparison of three attack scenarios: Baseline, DoS, and FDI conditions across the same network layout, timing structure, and packet-capture method. Because the same communication path was kept in all three scenarios, the differences in the results are more likely to be caused by the attack condition. This makes the comparison more reliable and easier to interpret.

Another contribution is the clear comparison between availability-based and integrity-based attacks. The DoS scenario had the strongest effect on timing behavior. It increased mean latency, wide jitter variation, and produced stronger short-term IPDV fluctuation. The FDI scenario also affected timing in some parts of the analysis, but the effect remained smaller than in the DoS case. This difference is useful because it shows that timing-based metrics respond more clearly to communication disruption than to attacks that mainly alter transmitted data values.

The threshold comparison based on baseline statistics is another practical contribution of the project. It provided a useful comparison tool that supported the same pattern seen in the boxplots and time-series plots. Overall the contribution of the project is practical rather than theoretical. It offers a clear experiment, a repeatable workflow, and a realistic comparison of how two common attack types affect smart grid communication performance.

The results also suggest that smart grid monitoring should be able to detect both stronger timing disruption in DoS conditions and smaller timing changes that may appear during FDI activity.

6.5 Future Work

This project can be extended in several practical ways. One possible approach is to test additional attack types within the same environment. In this study, the focus was on DoS and FDI attacks, which represent communication disruption and data manipulation. Future work might include replay attacks, man-in-the-middle attack, or a hybrid attack combining DoS and FDI, as it could show how timing disruption and data manipulation interact when both attacks occur simultaneously.

Another possible extension is to increase the scale of the simulation. The current project used a controlled setup, which made it easier to compare results between scenarios. However, a larger topology with more substations, additional communication paths, and more realistic traffic conditions may provide additional insight. It would help show whether the same timing patterns still appear when the system becomes more complex. In addition, an extended environment setup would make the results more representative of real smart grid environments, which could have practical consequences.

The analysis could also be developed further. In this study, statistical summaries, boxplots, threshold comparisons, and time-series plots were sufficient to highlight clear differences between scenarios. In future work, formal statistical testing could be added to confirm that the differences remain consistent across baseline, DoS, and FDI. Machine learning techniques could also be applied to the collected network metrics to support automated anomaly detection in smart grid communication networks.

The project also points to a practical improvement in the experimental workflow. In this study, all runs followed the same timing structure and were repeated carefully under controlled conditions. In future work, parts of the process, such as environment setup, data capture, and post-processing, could be automated while keeping the same overall logic. This would reduce manual effort and make it easier to handle larger sets of experiments. A further step would be to examine whether the baseline ranges and threshold comparisons used here could support a lightweight anomaly-detection approach under controlled smart grid conditions. In this way, the project provides a useful starting point for future work in communication monitoring, anomaly detection, and smart grid resilience.

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A

Project Artefact

The artefact is provided as a zip file. This artefact includes the README file, experiment scripts, processed CSV files, and the final figures used in the analysis.

Artefact OneDrive link: Download Project Artefact

Word count metrics

NUC Bachelor Project Word Count:

Total Sum count: 15933 Words in text: 15615 Words in headers: 135 Words outside text (captions, etc.): 176 Number of headers: 49 Number of floats/tables/figures: 19 Number of math inlines: 5 Number of math displayed: 2
(errors:3) NOTE: References are excluded.